accordance with one of the *canonical* forms of Chapter 2 for transfer functions. For example, in observer canonical form we would have

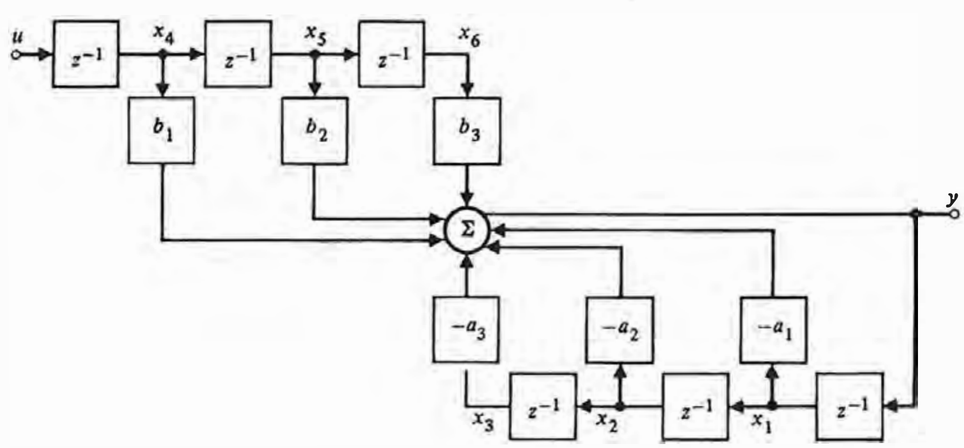
$$\mathbf{\Phi} = \begin{bmatrix} -a_1 & 1 & 0 \\ -a_2 & 0 & 1 \\ -a_3 & 0 & 0 \end{bmatrix} \quad \mathbf{\Gamma} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, \qquad \mathbf{H} = [1 \quad 0 \quad 0], \tag{12.38}$$

and we see immediately that these matrices are just functions of the $\boldsymbol{\theta}$ of Eq. (12.36), and the equivalence of $\boldsymbol{\theta}_1$ to $\boldsymbol{\theta}$ is obvious.

But now let us repeat, in this context, the comments made earlier about the difference between identification for control and modelling for science. In taking the canonical form represented by Eq. (12.38), we have almost surely scrambled into the a_i and b_i a jumble of the *physical* parameters of masses, lengths, spring constants, coefficients of friction, and so on. Although the *physical* nature of the problem can be best described and understood in terms of these numbers, it is the a_i and b_i that best serve our purpose of control, and it is with these that we will be concerned here.

Before leaving the matter of canonical forms, it is appropriate at this point to introduce another form called, in digital signal processing, the ARMA[4] model; this model has one feature especially appropriate for identification. A block diagram of the ARMA realization of $G(z)$ is shown in Fig. 12.6. Elementary calculations left to the reader will show that the transfer function of this block

Figure 12.6
Block diagram of the ARMA realization of $G(z)$



4 ARMA, an acronym for AutoRegressive Moving Average, comes from study of random processes generated by white noise filtered through the transfer function $G(z)$ of Eq. (12.35). We will have more to say about this after the element of randomness is introduced.

parameters of the ARMA model

diagram is given by Eq. (12.35). The state equations are more interesting. Here we find six states and the matrices

$$\mathbf{\Phi} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & b_1 & b_2 & b_3 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{\Gamma} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{H} = [\ -a_1 \quad -a_2 \quad -a_3 \quad b_1 \quad b_2 \quad b_3\ ]. \tag{12.39}$$

From the point of view of state-space analysis, the system described by Eq. (12.39) is seen to have six states to describe a third-order transfer function and thus to be "nonminimal." In fact, the $\mathbf{\Phi}$ matrix can be shown to have three poles at $z = 0$ that are *not observable* for any values of a_i or b_i. However, the system does have one remarkable property: The state is given by

$$\mathbf{x}(k) = [y(k-1) \quad y(k-2) \quad y(k-3) \quad u(k-1) \quad u(k-2) \quad u(k-3)]^T. \tag{12.40}$$

In other words, the state is exactly given by the past inputs and outputs so that, if we have the set of $[u(k)]$ and $[y(k)]$, we have the state also, since it is merely a listing of six members of the set.

All the action, as it were, takes place in the output equation, which is

$$\begin{aligned} y(k) &= \mathbf{Hx}(k) \\ &= -a_1 y(k-1) - a_2 y(k-2) - a_3 y(k-3) \\ &\quad + b_1 u(k-1) + b_2 u(k-2) + b_3 u(k-3). \end{aligned} \tag{12.41}$$

There is no need to carry any other equation along because the state equations are trivially related to this output equation. We will use the ARMA model in some of our later formulations for identification.

Thus we conclude that within the class of discrete parametric models we wish to select a model that has the fewest number of parameters and yet will be equivalent to the assumed plant description. A model whose parameters are uniquely determined by the observed data is highly desirable; a model that will make subsequent control design simple is also often selected.

12.3.2 Error Definition

Having selected the class of models described by our assumed plant description, we now turn to the techniques for selecting the particular estimate, $\hat{\boldsymbol{\theta}}$, that best represents the given data. For this we require some idea of goodness of fit of a proposed value of $\boldsymbol{\theta}$ to the true $\boldsymbol{\theta}^{\circ}$. Because, by the very nature of the problem, $\boldsymbol{\theta}^{\circ}$ is unknown, it is unrealistic to define a direct parameter error between $\boldsymbol{\theta}$ and $\boldsymbol{\theta}^{\circ}$. We must define the error in a way that can be computed from $[u(k)]$ and $[y(k)]$.

Three definitions that have been proposed and studied extensively are *equation error*, output error, and *prediction error*.

Equation Error

For the equation error, we need complete equations of motion as given, for example, by a state-variable description. To be just a bit general for the moment, suppose we have a nonlinear continuous time description with parameter vector $\boldsymbol{\theta}$, which we can write as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}; \boldsymbol{\theta}).$$

We assume, first, that we know the form of the vector functions $\mathbf{f}$ but not the particular parameters $\boldsymbol{\theta}^{\circ}$ that describe the plant. We assume, second, that we are able to measure not only the controls $\mathbf{u}$ but also the state $\mathbf{x}$ and the state derivative $\dot{\mathbf{x}}$. We thus know *everything* about the equations but the particular parameter values. We can, therefore, form an error comprised of the extent to which these equations of motion fail to be true for a specific value of $\boldsymbol{\theta}$ when used with the specific actual data $\mathbf{x}_a$, $\dot{\mathbf{x}}_a$, and $\mathbf{u}_a$. We write

$$\dot{\mathbf{x}}_a - \mathbf{f}(\mathbf{x}_a, \mathbf{u}_a; \boldsymbol{\theta}) = \mathbf{e}(t; \boldsymbol{\theta}),$$

equation error

and we assume that there is a vector of true parameters $\boldsymbol{\theta}^{\circ}$ such that $\mathbf{e}(t; \boldsymbol{\theta}^{\circ}) = 0$. The vector $\mathbf{e}(t; \boldsymbol{\theta})$ is defined as the **equation error**. The idea is that we would form some non-negative function of the error such as

$$\mathcal{J}(\boldsymbol{\theta}) = \int_0^T \mathbf{e}^T(t, \boldsymbol{\theta})\,\mathbf{e}(t, \boldsymbol{\theta})\,dt, \tag{12.42}$$

and search over $\boldsymbol{\theta}$ until we find $\hat{\boldsymbol{\theta}}$ such that $\mathcal{J}(\boldsymbol{\theta})$ is a minimum, at which time we will have a parameter set $\hat{\boldsymbol{\theta}}$ that is an estimate of $\boldsymbol{\theta}^{\circ}$. If we had an exact set of equations and selected a unique parameterization, then only one parameter set will make $\mathbf{e}(t; \boldsymbol{\theta}^{\circ}) = \mathbf{0}$, and so we will have $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}^{\circ}$. If noise is present in the equations of motion, then the error will not be zero at $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}^{\circ}$, but $\mathcal{J}$ will be minimized at that point.[5]

The assumption that we have enough sensors to measure the total state and all state derivatives is a strong assumption and often not realistic in continuous model identification. However, in discrete linear models there is one case where it is immediate, and that is the case of an ARMA model. The reason for this is not hard to find: In an ARMA model the state is no more than recent values of

5 Methods to search for the minimizing $\hat{\boldsymbol{\theta}}$ are the subject of nonlinear programming, which is discussed in Luenberger (1973). We give a brief discussion later in this chapter connected with maximum likelihood estimates.

input and output! To be explicit about it, let us write the linear, discrete model equation error, which is

$$\mathbf{x}_a(k+1) - \mathbf{\Phi}\mathbf{x}_a(k) - \mathbf{\Gamma}u_a(k) = \mathbf{e}(k;\boldsymbol{\theta}), \tag{12.43}$$

where $\mathbf{\Phi}$ and $\mathbf{\Gamma}$ are functions of the parameters $\boldsymbol{\theta}$. Now let us substitute the values from Eq. (12.39), the ARMA model

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \\ x_4(k+1) \\ x_5(k+1) \\ x_6(k+1) \end{bmatrix} - \begin{bmatrix} -a_1 & -a_2 & -a_3 & b_1 & b_2 & b_3 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \\ x_4(k) \\ x_5(k) \\ x_6(k) \end{bmatrix}$$

$$- \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} u(k) = \begin{bmatrix} e_1(k) \\ e_2(k) \\ e_3(k) \\ e_4(k) \\ e_5(k) \\ e_6(k) \end{bmatrix}. \tag{12.44}$$

When we make the substitution from Eq. (12.40) we find that, for any $\boldsymbol{\theta}$ (which is to say, for any values of a_i and b_i) the elements of equation error are *all zero* except e_1, and this element of error is given by

$$\begin{aligned} x_1(k+1) &+ a_1 x_1(k) + a_2 x_2(k) + a_3 x_3(k) \\ &- b_1 x_4(k) - b_2 x_5(k) - b_3 x_6(k) = e_1(k;\boldsymbol{\theta}) \end{aligned}$$

or

$$\begin{aligned} &y_a(k) + a_1 y_a(k-1) + a_2 y_a(k-2) + a_3 y_a(k-3) \\ &\quad - b_1 u_a(k-1) - b_2 u_a(k-2) - b_3 u_a(k-3) = e_1(k;\boldsymbol{\theta}). \end{aligned} \tag{12.45}$$

The performance measure Eq. (12.42) becomes

$$\mathcal{J}(\boldsymbol{\theta}) = \sum_{k=0}^{N} e_1^2(k;\boldsymbol{\theta}). \tag{12.46}$$

Again, we place the subscript a on the observed data to emphasize the fact that these are *actual* data that were produced via the plant with (we assume) parameter values $\boldsymbol{\theta}^\circ = [a_1^\circ\ a_2^\circ\ a_3^\circ\ b_1^\circ\ b_2^\circ\ b_3^\circ]$, and in Eq. (12.45) and Eq. (12.46) an error results because $\boldsymbol{\theta}$ differs from $\boldsymbol{\theta}^\circ$.

Output Error

As we have seen, the general case of equation error requires measurement of all elements of state and state derivatives. In order to avoid these assumptions a criterion based on output error can be used. This error is formulated in Fig. 12.7.

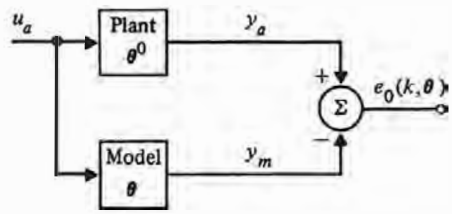


Figure 12.7
Block diagram showing the formulation of output error

Here we see that no attempt is made to measure the entire state of the plant, but rather the estimated parameter $\boldsymbol{\theta}$ is used in a model to produce the model output $y_m(k)$, which is a function, therefore, of $\boldsymbol{\theta}$ in the same way that the actual output is a function of the true parameter value $\boldsymbol{\theta}^{\circ}$. In the scalar output case the output error e_o will be a scalar and we can form the criterion function

$$\mathcal{J}(\boldsymbol{\theta}) = \sum_{k=0}^{N} e_o^2(k; \boldsymbol{\theta}) \tag{12.47}$$

and search for that $\hat{\boldsymbol{\theta}}$ which makes $\mathcal{J}(\hat{\boldsymbol{\theta}})$ as small as possible. To illustrate the difference between output error and equation error in one case, consider again the ARMA model for which we have

$$\begin{aligned} y_m(k) = &-a_1 y_m(k-1) - a_2 y_m(k-2) - a_3 y_m(k-3) \\ &+ b_1 u_a(k-1) + b_2 u_a(k-2) + b_3 u_a(k-3) \end{aligned}$$

output error

and then the output error

$$\begin{aligned} e_o(k; \boldsymbol{\theta}) &= y_a(k) - y_m(k) \\ &= y_a(k) + a_1 y_m(k-1) + a_2 y_m(k-2) + a_3 y_m(k-3) \\ &\quad -b_1 u_a(k-1) - b_2 u_a(k-2) - b_3 u_a(k-3). \end{aligned} \tag{12.48}$$

If we compare Eq. (12.48) with Eq. (12.45) we see that the equation error formulation has past values of the *actual* output, and the output error formulation uses past values of the *model* output. Presumably the equation error is in some way better because it takes more knowledge into account, but we are not well situated to study the matter at this point.

Prediction Error

The third approach to developing an error signal by which a parameter search can be structured is the prediction error. When we discussed observers in Chapter 8 we made the point that a simple model is not a good basis for an observer because the equations of motion of the errors cannot be controlled. When we add random effects into our data-collection scheme, as we are about to do, much the same complaint can be raised about the generation of output errors. Instead of working with a simple model, with output $y_m(k)$, we are led to consider an output *predictor* that, within the confines of known structures, will do the best possible job of *predicting* $y(k)$ based on previous observations. We will see that in one special circumstance the ARMA model error as given by Eq. (12.45) has the least prediction error, but this is not true in general. Following our development of the observer, we might suspect that a *corrected* model would be appropriate, but we must delay any results in that area until further developments. For the moment, and for the purpose of displaying the output prediction error formulation, we have the situation pictured in Fig. 12.8.

prediction error

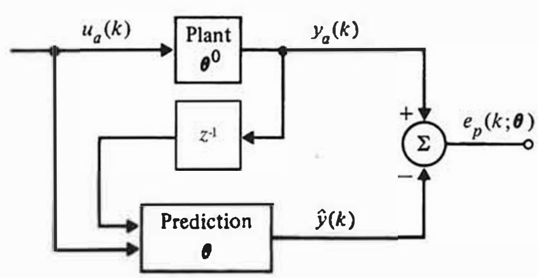
12.4 Deterministic Estimation

Having defined an error that depends on $\boldsymbol{\theta}$, we can formulate a performance criterion and search for that $\hat{\boldsymbol{\theta}}$ which is the best estimate of $\boldsymbol{\theta}^\circ$ within the context of the defined performance. To illustrate this use of error we have already defined a $\mathcal{J}(\boldsymbol{\theta})$ in Eq. (12.42), Eq. (12.46), and Eq. (12.47), in each case a sum of squares. The choice of a performance criterion is guided both by the computational difficulties it imposes in the effort to obtain $\hat{\boldsymbol{\theta}}$ and by the properties of the estimate that results, that is, by how hard $\hat{\boldsymbol{\theta}}$ is to find and how good it is when found. Among the criteria most widely used are:

1. least-squares estimate (LS),
2. best linear unbiased estimate (BLUE), and
3. maximum likelihood estimate (MLE)

The last two criteria require a stochastic element to be introduced into the model, and our understanding of least squares is enhanced by a stochastic perspective. However, before we introduce random noise into the picture, it is informative to consider deterministic least squares. Thus we will discuss least squares first, then introduce random noise and discuss the formulation of random errors, and finally define and discuss the BLUE and the MLE, their calculation and their properties.

Figure 12.8
Block diagram showing the generation of prediction error



12.4.1 Least Squares

To begin our discussion of least squares, we will take the ARMA model and equation error, which leads us to Eq. (12.45), repeated below for the nth-order case, where the subscript a is understood but omitted

$$\begin{aligned} y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) \\ - b_1 u(k-1) - \cdots - b_n u(k-n) = e(k;\ \boldsymbol{\theta}). \end{aligned} \tag{12.49}$$

We assume that we observe the set of outputs and inputs

$$\{-y(0), -y(1), \ldots, -y(N), u(0), u(1), \ldots, u(N)\}$$

and wish to compute values for

$$\boldsymbol{\theta} = [a_1 \cdots a_n\ b_1 \cdots b_n]^T,$$

which will best fit the observed data. Because $y(k)$ depends on past data back to n periods earlier, the first error we can form is $e(n;\ \boldsymbol{\theta})$. Suppose we define the *vector* of errors by writing Eq. (12.49) over and over for $k = n, n+1, \ldots, N$. The results would be

$$\begin{aligned} y(n) &= \boldsymbol{\phi}^T(n)\boldsymbol{\theta} + e(n;\ \boldsymbol{\theta}), \\ y(n+1) &= \boldsymbol{\phi}^T(n+1)\boldsymbol{\theta} + e(n+1;\ \boldsymbol{\theta}), \\ &\ \vdots \\ y(N) &= \boldsymbol{\phi}^T(N)\boldsymbol{\theta} + e(N;\ \boldsymbol{\theta}), \end{aligned} \tag{12.50}$$

where we have used the fact that the state of the ARMA model is[6]

$$\boldsymbol{\phi}(k) = [-y(k-1) - y(k-2) \cdots u(k-1) \cdots u(k-n)]^T.$$

To make the error even more compact, we introduce another level of matrix notation and define

$$\begin{aligned} \mathbf{Y}(N) &= [y(n)\ \cdots\ y(N)]^T, \\ \boldsymbol{\Phi}(N) &= [\boldsymbol{\phi}(n)\ \boldsymbol{\phi}(n+1)\ \cdots\ \boldsymbol{\phi}(N)]^T, \\ \boldsymbol{\epsilon}(N;\ \boldsymbol{\theta}) &= [e(n)\ \cdots\ e(N)]^T, \\ \boldsymbol{\theta} &= [a_1\ \cdots\ a_n\ b_1\ \cdots\ b_n]^T. \end{aligned} \tag{12.51}$$

Note that $\boldsymbol{\Phi}(N)$ is a *matrix* with $2n$ columns and $N - n + 1$ rows. In terms of these, we can write the equation errors as

$$\mathbf{Y} = \boldsymbol{\Phi}\boldsymbol{\theta} + \boldsymbol{\epsilon}(N;\ \boldsymbol{\theta}). \tag{12.52}$$

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6 We use $\boldsymbol{\phi}$ for the state here rather than $\mathbf{x}$ as in Eq. (12.44) because we will soon be developing equations for the evolution of $\boldsymbol{\theta}(k)$, estimates of the parameters, which will be the state of our identification dynamic system.

Least squares is a prescription that one should take that value of $\boldsymbol{\theta}$ which makes the sum of the squares of the $e(k)$ as small as possible. In terms of Eq. (12.50), we define

$$\mathcal{J}(\boldsymbol{\theta}) = \sum_{k=n}^{N} e^2(k;\, \boldsymbol{\theta}), \tag{12.53}$$

and in terms of Eq. (12.52), this is

$$\mathcal{J}(\boldsymbol{\theta}) = \boldsymbol{\epsilon}^T(N;\, \boldsymbol{\theta})\boldsymbol{\epsilon}(N;\, \boldsymbol{\theta}). \tag{12.54}$$

We want to find $\hat{\boldsymbol{\theta}}_{\mathrm{LS}}$, the least-squares estimate of $\boldsymbol{\theta}^0$, which is that $\boldsymbol{\theta}$ having the property

$$\mathcal{J}(\hat{\boldsymbol{\theta}}_{\mathrm{LS}}) \le \mathcal{J}(\boldsymbol{\theta}). \tag{12.55}$$

But $\mathcal{J}(\boldsymbol{\theta})$ is a quadratic function of the $2n$ parameters in $\boldsymbol{\theta}$, and from calculus we take the result that a necessary condition on $\hat{\boldsymbol{\theta}}_{\mathrm{LS}}$ is that the partial derivatives of $\mathcal{J}$ with respect to $\boldsymbol{\theta}$ at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_{\mathrm{LS}}$ should be zero. This we do as follows

$$\begin{aligned}\mathcal{J}_{\boldsymbol{\theta}} &= \boldsymbol{\epsilon}^T\boldsymbol{\epsilon}\\ &= (\mathbf{Y} - \boldsymbol{\Phi}\boldsymbol{\theta})^T(\mathbf{Y} - \boldsymbol{\Phi}\boldsymbol{\theta})\\ &= \mathbf{Y}^T\mathbf{Y} - \boldsymbol{\theta}^T\boldsymbol{\Phi}^T\mathbf{Y} - \mathbf{Y}^T\boldsymbol{\Phi}\boldsymbol{\theta} + \boldsymbol{\theta}^T\boldsymbol{\Phi}^T\boldsymbol{\Phi}\boldsymbol{\theta};\end{aligned}$$

and applying the rules developed above for derivatives of scalars with respect to vectors[7] we obtain

$$\mathcal{J}_{\boldsymbol{\theta}} = \left[\frac{\partial \mathcal{J}}{\partial \boldsymbol{\theta}_i}\right] = -2\mathbf{Y}^T\boldsymbol{\Phi} + 2\boldsymbol{\theta}^T\boldsymbol{\Phi}^T\boldsymbol{\Phi}. \tag{12.56}$$

If we take the transpose of Eq. (12.56) and let $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_{\mathrm{LS}}$, we must get zero; thus

$$\boldsymbol{\Phi}^T\boldsymbol{\Phi}\hat{\boldsymbol{\theta}}_{\mathrm{LS}} = \boldsymbol{\Phi}^T\mathbf{Y}. \tag{12.57}$$

These equations are called the *normal equations* of the problem, and their solution will provide us with the least-squares estimate $\hat{\boldsymbol{\theta}}_{\mathrm{LS}}$.

Do the equations have a unique solution? The answer depends mainly on how $\boldsymbol{\theta}$ was selected and what input signals $\{u(k)\}$ were used. Recall that earlier we saw that a general third-order state model had fifteen parameters, but that only six of these were needed to completely describe the input–output dependency. If we stayed with the fifteen-element $\boldsymbol{\theta}$, the resulting normal equations could not have a unique solution. To obtain a unique parameter set, we must select a canonical form having a minimal number of parameters, such as the observer or ARMA forms. By way of definition, a parameter $\boldsymbol{\theta}$ having the property that one

7 The reader who has not done so before should write out $\mathbf{a}^T\mathbf{Q}\mathbf{a}$ for a 3×3 case and verify that $\partial\mathbf{a}^T\mathbf{Q}\mathbf{a}/\partial\mathbf{a} = 2\mathbf{a}^T\mathbf{Q}$.

and only one value of $\boldsymbol{\theta}$ makes $\mathcal{J}(\boldsymbol{\theta})$ a minimum is said to be "identifiable." Two parameters having the property that $\mathcal{J}(\boldsymbol{\theta}_1) = \mathcal{J}(\boldsymbol{\theta}_2)$ are said to be "equivalent."

As to the selection of the inputs $u(k)$, let us consider an absurd case. Suppose $u(k) \equiv c$ for all k — a step function input. Now suppose we look at Eq. (12.50) again for the third-order case to be specific. The errors are

$$\begin{aligned} y(3) &= -a_1 y(2) - a_2 y(1) - a_3 y(0) + b_1 c + b_2 c + b_3 c + e(3), \\ &\vdots \\ y(N) &= -a_1 y(N_1) - a_2 (N-2) - a_3 (N-3) + b_1 c + b_2 c + b_3 c + e(N). \end{aligned} \tag{12.58}$$

It is obvious that in Eq. (12.58) the parameters b_1, b_2, and b_3 always appear as the sum $b_1 + b_2 + b_3$ and that separation of them is not possible when a constant input is used. Somehow the constant u fails to "excite" all the dynamics of the plant. This problem has been studied extensively, and the property of "persistently exciting" has been defined to describe a sequence $\{u(k)\}$ that fluctuates enough to avoid the possibility that only linear combinations of elements of $\boldsymbol{\theta}$ will show up in the error and hence in the normal equations [Ljung(1987)]. Without being more specific at this point, we can say that an input is *persistently exciting of order n* if the lower right $(n \times n)$-matrix component of $\boldsymbol{\Phi}^T\boldsymbol{\Phi}$ [which depends only on $\{u(k)\}$] is nonsingular. It can be shown that a signal is persistently exciting of order n if its discrete spectrum has at least n nonzero points over the range $0 \leq \omega T < \pi$. White noise and a pseudo-random binary signal are examples of frequently used persistently exciting input signals.

For the moment, then, we will *assume* that the $u(k)$ are persistently exciting and that the $\boldsymbol{\theta}$ are identifiable and consequently that $\boldsymbol{\Phi}^T\boldsymbol{\Phi}$ is nonsingular. We can then write the explicit solution

$$\hat{\boldsymbol{\theta}}_{LS} = (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{Y}. \tag{12.59}$$

It should be especially noted that although we are here mainly interested in identification of parameters to describe dynamic systems, the solution Eq. (12.59) derives entirely from the error equations Eq. (12.52) and the sum of squares criterion Eq. (12.54). Least squares is used for all manner of curve fitting, including nonlinear least squares when the error is a nonlinear function of the parameters. Numerical methods for solving for the least-squares solution to Eq. (12.57) without ever explicitly forming the product $\boldsymbol{\Phi}^T\boldsymbol{\Phi}$ have been extensively studied [see Golub (1965) and Strang (1976)].

The performance measure Eq. (12.53) is essentially based on the view that all the errors are equally important. This is not necessarily so, and a very simple modification can take account of known differences in the errors. We might know, for example, that data taken later in the experiment were much more in error than data taken early on, and it would seem reasonable to weight the errors accordingly. Such a scheme is referred to as weighted least squares and is based on the performance criterion

$$\mathcal{J}(\boldsymbol{\theta}) = \sum_{k=n}^{N} w(k)e^2(k;\ \boldsymbol{\theta}) = \boldsymbol{\epsilon}^T \mathbf{W} \boldsymbol{\epsilon}. \tag{12.60}$$

In Eq. (12.60) we take the weighting function $w(k)$ to be positive and, presumably, to be small where the errors are expected to be large, and vice versa. In any event, derivation of the normal equations from Eq. (12.60) follows at once and gives

$$\boldsymbol{\Phi}^T \mathbf{W} \boldsymbol{\Phi} \hat{\boldsymbol{\theta}}_{\text{WLS}} = \Phi^T \mathbf{W} \mathbf{Y},$$

and, subject to the coefficient matrix being nonsingular, we have

$$\hat{\boldsymbol{\theta}}_{\text{WLS}} = (\boldsymbol{\Phi}^T \mathbf{W} \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^T \mathbf{W} \mathbf{Y}. \tag{12.61}$$

We note that Eq. (12.61) reduces to ordinary least squares when $\mathbf{W} = \mathbf{I}$, the identity matrix. Another common choice for $w(k)$ in Eq. (12.60) is $w(k) = (1 - \gamma)\gamma^{N-k}$ for $\gamma < 1$. This choice weights the recent (k near N) observations more than the past (k near n) ones and corresponds to a first-order filter operating on the squared error. The factor $1 - \gamma$ causes the gain of the equivalent filter to be 1 for constant errors. As γ nears 1, the filter memory becomes long, and noise effects are reduced; whereas for smaller γ, the memory is short, and the estimate can track changes that can occur in $\boldsymbol{\theta}$ if the computation is done over and over as N increases. The past is weighted geometrically with weighting γ, but a rough estimate of the memory length is given by $1/(1 - \gamma)$; so, for example, a $\gamma = 0.99$ corresponds to a memory of about 100 samples. The choice is a compromise between a short memory which permits tracking changing parameters and a long memory which incorporates a lot of averaging and reduces the noise variance.

12.4.2 Recursive Least Squares

The weighted least-squares calculation for $\hat{\boldsymbol{\theta}}_{\text{WLS}}$ given in Eq. (12.61) is referred to as a "batch" calculation, because by the definition of the several entries, the formula presumes that one has a batch of data of length N from which the matrices $\mathbf{Y}$ and $\boldsymbol{\Phi}$ are composed according to the definitions in Eq. (12.51), and from which, with the addition of the weighting matrix $\mathbf{W}$, the normal equations are solved. There are times when the data are acquired sequentially rather than in a batch, and other times when one wishes to examine the nature of the solution as more data are included to see whether perhaps some improvement in the parameter estimates continues to be made or whether any surprises occur such as a sudden change in $\boldsymbol{\theta}$ or a persistent drift in one or more of the parameters. In short, one wishes sometimes to do a visual or experimental examination of the new estimates as one or several more data points are included in the computed values of $\hat{\boldsymbol{\theta}}_{\text{WLS}}$.

The equations of Eq. (12.61) can be put into a form for sequential processing of the type described. We begin with Eq. (12.61) as solved for N data points and consider the consequences of taking one more observation. We need to consider the structure of $\mathbf{\Phi}^T\mathbf{W}\mathbf{\Phi}$ and $\mathbf{\Phi}^T\mathbf{W}\mathbf{Y}$ as one more datum is added. Consider first $\mathbf{\Phi}^T\mathbf{W}\mathbf{\Phi}$. To be specific about the weights, we will assume $w = a\gamma^{N-k}$. Then, if $a = 1$ and $\gamma = 1$, we have ordinary least squares; and if $a = 1 - \gamma$, we have exponentially weighted least squares. From Eq. (12.51) we have, for data up to time $N + 1$

$$\mathbf{\Phi}^T = [\boldsymbol{\phi}(n) \cdots \boldsymbol{\phi}(N)\ \boldsymbol{\phi}(N+1)]$$

and

$$\mathbf{\Phi}^T\mathbf{W}\mathbf{\Phi} = \sum_{k=n}^{N+1} \boldsymbol{\phi}(k)w(k)\boldsymbol{\phi}^T(k) = \sum_{k=n}^{N+1} \boldsymbol{\phi}(k)a\gamma^{N+1-k}\boldsymbol{\phi}^T(k),$$

which can be written in two terms as

$$\begin{aligned}\mathbf{\Phi}^T\mathbf{W}\mathbf{\Phi} &= \sum_{k=n}^{N} \boldsymbol{\phi}(k)a\gamma\gamma^{N-k}\boldsymbol{\phi}^T(k) + \boldsymbol{\phi}(N+1)a\boldsymbol{\phi}^T(N+1) \\ &= \gamma\boldsymbol{\phi}^T(N)\mathbf{W}(N)\mathbf{\Phi}(N) + \boldsymbol{\phi}(N+1)a\boldsymbol{\phi}^T(N+1). \qquad (12.62)\end{aligned}$$

From the solution Eq. (12.61) we see that the inverse of the matrix in Eq. (12.62) will be required,[8] and for convenience and by convention we *define* the $2n \times 2n$ matrix $\mathbf{P}$ as

$$\mathbf{P}(N+1) = [\mathbf{\Phi}^T(N+1)\mathbf{W}\mathbf{\Phi}(N+1)]^{-1}. \qquad (12.63)$$

Then we see that Eq. (12.62) can be written as

$$\mathbf{P}(N+1) = [\gamma\mathbf{P}^{-1}(N) + \boldsymbol{\phi}(N+1)a\boldsymbol{\phi}^T(N+1)]^{-1}, \qquad (12.64)$$

and we need the inverse of a sum of two matrices. This is a well-known problem, and a formula attributed to Householder (1964) known as the *matrix inversion lemma* is

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{D}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{D}\mathbf{A}^{-1}. \qquad (12.65)$$

To apply Eq. (12.65) to Eq. (12.62) we make the associations

$$\begin{aligned}\mathbf{A} &= \gamma\mathbf{P}^{-1}(N), \\ \mathbf{B} &= \boldsymbol{\phi}(N+1) \equiv \boldsymbol{\phi}, \\ \mathbf{C} &= w(N+1) \equiv a, \\ \mathbf{D} &= \boldsymbol{\phi}^T(N+1) \equiv \boldsymbol{\phi}^T,\end{aligned}$$

8 We assume for this discussion the existence of the inverse which assumes inputs that are persistently exciting and parameters that are identifiable.

and we find at once that

$$\mathbf{P}(N+1) = \frac{\mathbf{P}(N)}{\gamma} - \frac{\mathbf{P}(N)}{\gamma}\boldsymbol{\phi}\left(\frac{1}{a} + \boldsymbol{\phi}^T\frac{\mathbf{P}(N)}{\gamma}\boldsymbol{\phi}\right)^{-1}\boldsymbol{\phi}^T\frac{\mathbf{P}(N)}{\gamma}. \tag{12.66}$$

In the solution we also need $\boldsymbol{\Phi}^T\mathbf{WY}$, which we write as

$$\boldsymbol{\Phi}^T\mathbf{WY} = [\boldsymbol{\phi}(n)\cdots\boldsymbol{\phi}(N) \quad \boldsymbol{\phi}(N+1)]\begin{bmatrix} a\gamma^{N+1-n} & & & \bigcirc \\ & \ddots & & \\ & & a\gamma & \\ \bigcirc & & & a \end{bmatrix}\begin{bmatrix} y(n) \\ \vdots \\ y(N) \\ y(N+1) \end{bmatrix}, \tag{12.67}$$

which can be expressed in two terms as

$$\boldsymbol{\Phi}^T\mathbf{WY}(N+1) = \gamma\,\boldsymbol{\Phi}^T\mathbf{WY}(N) + \boldsymbol{\phi}(N+1)ay(N+1). \tag{12.68}$$

If we now substitute the expression for $\mathbf{P}(N+1)$ from Eq. (12.66) and for $\boldsymbol{\Phi}\mathbf{WY}(N+1)$ from Eq. (12.68) into Eq. (12.61), we find [letting $\mathbf{P}(N) = \mathbf{P}$, $\boldsymbol{\phi}(N+1) = \boldsymbol{\phi}$, and $y(N+1) = y$ for notational convenience]

$$\hat{\boldsymbol{\theta}}_{\text{WLS}}(N+1) = \left[\frac{\mathbf{P}}{\gamma} - \frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\left(\frac{1}{a} + \boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\right)^{-1}\boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\right][\gamma\boldsymbol{\Phi}^T\mathbf{WY}(N) + \boldsymbol{\phi}ay]. \tag{12.69}$$

When we multiply the factors in Eq. (12.69), we see that the term $\mathbf{P}\boldsymbol{\Phi}^T\mathbf{WY}(N) = \hat{\boldsymbol{\theta}}_{\text{WLS}}(N)$, so that Eq. (12.69) reduces to

$$\begin{aligned}\hat{\boldsymbol{\theta}}_{\text{WLS}}(N+1) = \hat{\boldsymbol{\theta}}_{\text{WLS}}(N) &+ \frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}ay - \frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\left(\frac{1}{a} + \boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\right)^{-1}\boldsymbol{\phi}^T\hat{\boldsymbol{\theta}}_{\text{WLS}} \\ &- \frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\left(\frac{1}{a} + \boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\right)^{-1}\boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}ay.\end{aligned} \tag{12.70}$$

If we now insert the identity

$$\left(\frac{1}{a} + \boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\right)^{-1}\left(\frac{1}{a} + \boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\right)$$

between the $\boldsymbol{\phi}$ and the a in the second term on the right of Eq. (12.70), we can combine the two terms which multiply y to reduce Eq. (12.70) to

$$\hat{\boldsymbol{\theta}}_{\text{WLS}}(N+1) = \hat{\boldsymbol{\theta}}_{\text{WLS}}(N) + \mathbf{L}(N+1)(y(N+1) - \boldsymbol{\phi}^T\hat{\boldsymbol{\theta}}_{\text{WLS}}(N)), \tag{12.71}$$

where we have defined

$$\mathbf{L}(N+1) = \frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\left(\frac{1}{a} + \frac{\boldsymbol{\phi}^T\mathbf{P}\boldsymbol{\phi}}{\gamma}\right)^{-1}. \tag{12.72}$$

Equations (12.66), (12.71), and (12.72) can be combined into a set of steps that constitute an algorithm for computing $\hat{\boldsymbol{\theta}}$ recursively. To collect these, we proceed as follows:

1. Select $a\ \gamma$, and N.
2. Comment: $a = \gamma = 1$ is ordinary least squares; $a = 1 - \gamma$ and $0 < \gamma < 1$ is exponentially weighted least squares.
3. Select initial values for $\mathbf{P}(N)$ and $\hat{\boldsymbol{\theta}}(N)$. *Comment:* See discussion below.
4. Collect $y(0), \ldots, y(N)$ and $u(0), \ldots, u(N)$ and form $\boldsymbol{\phi}^T(N+1)$.
5. Let $k \leftarrow N$.
6. $\mathbf{L}(k+1) \leftarrow \dfrac{\mathbf{P}(k)}{\gamma}\boldsymbol{\phi}(k+1)\left(\dfrac{1}{a} + \boldsymbol{\phi}^T(k+1)\dfrac{\mathbf{P}(k)}{\gamma}\boldsymbol{\phi}(k+1)\right)^{-1}$
7. Collect $y(k+1)$and $u(k+1)$.
8. $\hat{\boldsymbol{\theta}}(k+1) \leftarrow \hat{\boldsymbol{\theta}}(k) + \mathbf{L}(k+1)(y(k+1) - \boldsymbol{\phi}^T(k+1)\hat{\boldsymbol{\theta}}(k))$
9. $\mathbf{P}(k+1) \leftarrow \dfrac{1}{\gamma}[\mathbf{I} - \mathbf{L}(k+1)\boldsymbol{\phi}^T(k+1)]\mathbf{P}(k)$
10. Form $\boldsymbol{\phi}(k+2)$.
11. Let $k \leftarrow k+1$.
12. Go to Step 6.

Especially pleasing is the form of Step 8, the "update" formula for the next value of the estimate. We see that the term $\boldsymbol{\phi}^T\hat{\boldsymbol{\theta}}(N)$ is the output to be expected at the time $N+1$ based on the previous data, $\boldsymbol{\phi}(N+1)$, and the previous estimate, $\hat{\boldsymbol{\theta}}(N)$. Thus the next estimate of $\boldsymbol{\theta}$ is given by the old estimate corrected by a term linear in the error between the observed output, $y(N+1)$, and the predicted output, $\boldsymbol{\phi}^T\hat{\boldsymbol{\theta}}(N)$. The gain of the correction, $\mathbf{L}(N+1)$, is given by Eq. (12.71) and Eq. (12.72). Note especially that in Eq. (12.71) no matrix inversion is required but only division by the scalar

$$\frac{1}{a} + \boldsymbol{\phi}^T\frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}.$$

However, one should not take the implication that Eq. (12.71) is without numerical difficulties, but their study is beyond the scope of this text.

We still have the question of initial conditions to resolve. Two possibilities are commonly recommended:

1. Collect a batch of $N > 2n$ data values and solve the batch formula Eq. (12.61) once for $\mathbf{P}(N)$, $\mathbf{L}(N+1)$, and $\hat{\boldsymbol{\theta}}(N)$, and enter these values at Step 3.
2. Set $\hat{\boldsymbol{\theta}}(N) = \mathbf{0}$, $\mathbf{P}(N) = \alpha\mathbf{I}$, where α is a large scalar. The suggestion has been made that an estimate of a suitable α is [Soderstrom, Ljung, and Gustavsson (1974)]

$$\alpha = (10)\frac{1}{N+1}\sum_{i=0}^{N} y^2(i).$$

The steps in the table update the least-squares estimate of the parameters $\boldsymbol{\theta}$ when one more pair of data points u and y are taken. With only modest effort we can extend these formulas to include the case of vector or multivariable observations wherein the data $\mathbf{y}(k)$ are a vector of p simultaneous observations. We assume that parameters $\boldsymbol{\theta}$ have been defined such that the system can be described by

$$\mathbf{y}(k) = \boldsymbol{\phi}^T(k)\boldsymbol{\theta} + \boldsymbol{\epsilon}(k;\ \boldsymbol{\theta}). \tag{12.73}$$

One such set of parameters is defined by the multivariable ARMA model

$$\mathbf{y}(k) = -\sum_{i=1}^{n} a_i \mathbf{y}(k-i) + \sum_{i=1}^{n} \mathbf{B}_i \mathbf{u}(k-i), \tag{12.74}$$

where the a_i are scalars, the $\mathbf{B}_i$ are $p \times m$ matrices, $\boldsymbol{\theta}$ is $(n+nmp) \times 1$, and the $\boldsymbol{\phi}(k)$ are now $(n+nmp) \times p$ matrices. If we define $\boldsymbol{\Phi}$, $\mathbf{Y}$, and $\boldsymbol{\epsilon}$ as in Eq. (12.51), the remainder of the batch formula development proceeds exactly as before, leading to Eq. (12.59) for the least-squares estimates and Eq. (12.61) for the weighted least-squares estimates with little more than a change in the definition of the elements in the equations. We need to modify Eq. (12.60) to $\mathcal{J} = \Sigma \boldsymbol{\epsilon}^T \mathbf{w} \boldsymbol{\epsilon}$, reflecting the fact that $\boldsymbol{\epsilon}(k)$ is now also a $p \times 1$ vector and the $\mathbf{w}(k)$ are $p \times p$ nonsingular matrices.

To compute the recursive estimate equations, we need to repeat Eq. (12.62) and the development following Eq. (12.62) with the new definitions. Only minor changes are required. For example, in Eq. (12.66) we must replace $1/a$ by $\mathbf{a}^{-1}$. The resulting equations are, in the format of Eq. (12.71)

$$\begin{aligned}
\mathbf{L}(N+1) &= \frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\left(\mathbf{a}^{-1} + \boldsymbol{\phi}^T \frac{\mathbf{P}}{\gamma}\boldsymbol{\phi}\right)^{-1}, && (a)\\
\mathbf{P}(N+1) &= \frac{1}{\gamma}(\mathbf{I} - \mathbf{L}(N+1)\boldsymbol{\phi}^T)\mathbf{P}, && (b)\\
\hat{\boldsymbol{\theta}}_{\text{WLS}}(N+1) &= \hat{\boldsymbol{\theta}}_{\text{WLS}} + \mathbf{L}(N+1)[\mathbf{y}(N+1) - \boldsymbol{\phi}^T\hat{\boldsymbol{\theta}}_{\text{WLS}}(N)]. && (c)
\end{aligned} \tag{12.75}$$

12.5 Stochastic Least Squares

Thus far we have presented the least-squares method with no comment about the possibility that the data might in fact be subject to random effects. Because such effects are very common and often are the best available vehicle for describing the differences between an ideal model and real-plant signal observations, it is essential that some account of such effects be included in our calculations. We will begin with an analysis of the most elementary of cases and add realism (and difficulties) as we go along. Appendix D provides a brief catalog of results we will need from probability, statistics, and stochastic processes.

As a start, we consider the case of a deterministic model with random errors in the data. We consider the equations that generate the data to be[9]

$$y(k) = \mathbf{a}^T\boldsymbol{\theta}^0 + v(k), \tag{12.76}$$

which, in matrix notation, becomes

$$\mathbf{Y} = \mathbf{A}\boldsymbol{\theta}^0 + \mathbf{V}. \tag{12.77}$$

The $v(k)$ in Eq. (12.76) are assumed to be random variables with zero mean (one can always subtract a known mean) and known covariance. In particular, we assume that the actual data are generated from Eq. (12.77) with $\boldsymbol{\theta} = \boldsymbol{\theta}^0$ and

$$\begin{aligned} \mathcal{E}v(k) &= 0, \\ \mathcal{E}v(k)v(j) &= \sigma^2\delta_{kj} = \begin{cases} 0 & (k \neq j) \\ \sigma^2 & (k = j) \end{cases} \end{aligned} \tag{12.78}$$

or

$$\mathcal{E}\mathbf{V}\mathbf{V}^T = \sigma^2\mathbf{I}.$$

As an example of the type considered here, suppose we have a physics experiment in which observations are made of the positions of a mass that moves without forces but with unknown initial position and velocity. We assume then that the motion will be a line, which may be written as

$$y(t) = a^0 + b^0 t + n(t). \tag{12.79}$$

If we take observations at times $t_0, t_1, t_2, \ldots, t_N$, then

$$\mathbf{Y} = [y(t_0)\ldots y(t_N)]^T, \quad \mathbf{A} = \begin{pmatrix} t_0 & t_1 & \cdots & t_N \\ 1 & 1 & \cdots & 1 \end{pmatrix}^T,$$

the unknown parameters are

$$\boldsymbol{\theta} = [a \quad b]^T,$$

and the noise is

$$v(k) = n(t_k).$$

We assume that the observations are without systematic bias—that is, the noise has zero average value—and that we can estimate the noise "intensity," or mean-square value, σ^2. We assume zero error in the clock times, t_k, so $\mathbf{A}$ is known.

The (stochastic) least-squares problem is to find $\hat{\boldsymbol{\theta}}$ which will minimize

$$\begin{aligned} \mathcal{J}(\boldsymbol{\theta}) &= (\mathbf{Y} - \mathbf{A}\boldsymbol{\theta})^T(\mathbf{Y} - \mathbf{A}\boldsymbol{\theta}) \\ &= \sum_{k=0}^{N} e^2(k;\, \boldsymbol{\theta}). \end{aligned} \tag{12.80}$$

9 There is no connection between the **a** in (12.76) and the a used as part of the weights in Section 12.4.

Note that the errors, $e(k;\ \boldsymbol{\theta})$, depend both on the random noise and on the choice of $\boldsymbol{\theta}$, and $e(k;\ \boldsymbol{\theta}^0) = v(k)$. Now the solution will be a random variable because the data, $\mathbf{Y}$, on which it is based, is random. However, for specific actual data, Eq. (12.80) represents the same quadratic function of $\boldsymbol{\theta}$ we have seen before; and the same form of the solution results, save only the substitution of $\mathbf{A}$ for $\boldsymbol{\Phi}$, namely

$$\hat{\boldsymbol{\theta}}_{LS} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y}. \tag{12.81}$$

Now, however, we should not expect to find zero for the sum of the errors given by Eq. (12.80), even if we should determine $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}^0$ exactly (which of course we won't). Because of the random effects, then, we must generalize our concepts of what constitutes a "good" estimate. We will use three features of a stochastic estimate. The first of these is consistency.

consistent estimate

An estimate $\hat{\boldsymbol{\theta}}$ of a parameter $\boldsymbol{\theta}^0$ is said to be *consistent* if, in the long run, the difference between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}^0$ becomes negligible. In statistics and probability, there are several formal ways by which one can define a negligible difference. For our purposes we will use the mean-square criterion, by which we measure the difference between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}^0$ by the sum of the squares of the parameter error, $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0$. We make explicit the dependence of $\hat{\boldsymbol{\theta}}$ on the length of the data by writing $\hat{\boldsymbol{\theta}}(N)$ and say that an estimate is consistent if [10]

$$\begin{aligned}\lim_{N\to\infty} \mathcal{E}(\hat{\boldsymbol{\theta}}(N) - \boldsymbol{\theta}^0)^T(\hat{\boldsymbol{\theta}}(N) - \boldsymbol{\theta}^0) &= 0,\\ \lim_{N\to\infty} \operatorname{tr}\mathcal{E}(\hat{\boldsymbol{\theta}}(N) - \boldsymbol{\theta}^0)(\hat{\boldsymbol{\theta}}(N) - \boldsymbol{\theta}^0)^T &= 0.\end{aligned} \tag{12.82}$$

If Eq. (12.82) is true, we say that $\hat{\boldsymbol{\theta}}(N)$ converges to $\boldsymbol{\theta}^0$ in the mean-square sense as N approaches infinity. The expression in Eq. (12.82) can be made more explicit in the case of the least-squares estimate of Eq. (12.81). We have

$$\begin{aligned}\hat{\boldsymbol{\theta}}_{\text{LS}} - \boldsymbol{\theta}^0 &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T(\mathbf{A}\boldsymbol{\theta}^0 + \mathbf{V}) - \boldsymbol{\theta}^0\\ &= \boldsymbol{\theta}^0 + (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{V} - \boldsymbol{\theta}^0\\ &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{V}\end{aligned}$$

and

$$\begin{aligned}\mathcal{E}(\hat{\boldsymbol{\theta}}_{\text{LS}} - \boldsymbol{\theta}^0)(\hat{\boldsymbol{\theta}}_{\text{LS}} - \boldsymbol{\theta}^0)^T &= \mathcal{E}\{(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{V}\mathbf{V}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\}\\ &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathcal{E}\mathbf{V}\mathbf{V}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\\ &= \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}.\end{aligned} \tag{12.83}$$

In making the reductions shown in the development of Eq. (12.83) we have used the facts that $\mathbf{A}$ is a known matrix (not random) and that $E\mathbf{V}\mathbf{V}^T = \sigma^2\mathbf{I}$.

10 The trace (tr) of a matrix $\mathbf{A}$ with elements a_{ij} is the sum of the diagonal elements, or trace $\mathbf{A} = \Sigma_{i=1}^n a_{ii}$.

Continuing then with consideration of the least-squares estimate, we see that $\hat{\boldsymbol{\theta}}_{\text{LS}}$ is consistent if

$$\lim_{N\to\infty} \operatorname{tr} \sigma^2(\mathbf{A}^T\mathbf{A})^{-1} = 0. \tag{12.84}$$

As an example of consistency, we consider the most simple of problems, the observation of a constant. Perhaps we have the mass of the earlier problem, but with zero velocity. Thus

$$y(k) = a^0 + v(k), \quad k = 0, 1, \dots, N, \tag{12.85}$$

and

$$\mathbf{A} = [1 \quad 1 \dots 1]^T, \quad \boldsymbol{\theta}^0 = a^0,$$
$$\mathbf{A}^T\mathbf{A} = \sum_{k=0}^{N} 1 = N + 1, \quad \mathbf{A}^T\mathbf{Y} = \sum_{k=0}^{N} y(k);$$

then

$$\hat{\boldsymbol{\theta}}_{\text{LS}}(N) = (N+1)^{-1} \sum_{k=0}^{N} y(k), \tag{12.86}$$

which is the sample average.

Now, if we apply Eq. (12.84), we find

$$\lim_{N\to\infty} \operatorname{tr} \sigma^2 (N+1)^{-1} = \lim_{N\to\infty} \frac{\sigma^2}{N+1} = 0, \tag{12.87}$$

and we conclude that this estimate Eq. (12.86) is a *consistent* estimate of a^0. If we keep taking observations according to Eq. (12.85) and calculating the sum according to Eq. (12.86), we will eventually have a value that differs from a^0 by a negligible amount in the mean-square sense.

The estimate given by Eq. (12.86) is a batch calculation. It can be informative to apply the recursive algorithm of Eq. (12.52) to this trivial case just to see how the equations will look. Suppose we agree to start the equations with one stage of "batch" on, say, two observations, $y(0)$ and $y(1)$. We thus have $a = \gamma = 1$, for least squares, and

$$\left.\begin{aligned} P(1) &= (A^TA)^{-1} = \tfrac{1}{2}, \\ \boldsymbol{\phi}(N) &= 1, \\ \hat{\boldsymbol{\theta}}(1) &= \tfrac{1}{2}(y(0)) + y(1)), \\ \mathbf{W} &= 1. \end{aligned}\right\} \quad \text{initial conditions} \tag{12.88}$$

The iteration in P is given by (because $\boldsymbol{\phi} \equiv 1$)

$$\begin{aligned} P(N+1) &= P(N) - \frac{P(N)(1)(1)P(N)}{1 + (1)P(N)(1)} \\ &= \frac{P(N)}{1 + P(N)}. \end{aligned}$$

Thus the entire iteration is given by the initial conditions of Eq. (12.88) plus:

$$
\begin{aligned}
&\textbf{1. } \text{Let } N = 1.\\
&\textbf{2. } P(N+1) = \frac{P(N)}{1+P(N)}.\\
&\textbf{3. } L(N+1) = \frac{P(N)}{1+P(N)}.\\
&\textbf{4. } \hat{\theta}(N+1) = \hat{\theta}(N) + L(N+1)(y(N+1) - \hat{\theta}(N)).\\
&\textbf{5. } \text{Let } N \text{ be replaced by } N+1.\\
&\textbf{6. } \text{Go to Step 2.}
\end{aligned}
\tag{12.89}
$$

The reader can verify that in this simple case, $\mathbf{P}(N) = 1/(N+1)$, and Step 4 of the recursive equations gives the same (consistent) estimate as the batch formula Eq. (4.60). We also note that the matrix $\mathbf{P}$ is proportional to the variance in the error of the parameter estimate as expressed in Eq. (12.83).

Although consistency is the first property one should expect of an estimate, it is, after all, an asymptotic property that describes a feature of $\hat{\boldsymbol{\theta}}$ as N grows without bound. A second property that can be evaluated is that of "bias." If $\hat{\boldsymbol{\theta}}$ is an estimate of $\boldsymbol{\theta}^0$, the bias in the estimate is the difference between the mean value of $\hat{\boldsymbol{\theta}}$ and the true value, $\boldsymbol{\theta}^0$. We have

$$\mathcal{E}\hat{\boldsymbol{\theta}}(N) - \boldsymbol{\theta}^0 = \mathbf{b} \tag{12.90}$$

If $\mathbf{b} = 0$ for all N, the estimate is said to be *unbiased*. The least-squares estimate given by Eq. (12.59) is unbiased, which we can prove by direct calculation as follows. If we return to the development of the mean-square error in $\hat{\boldsymbol{\theta}}_{\text{LS}}$ given by Eq. (12.83) we find that

$$\hat{\boldsymbol{\theta}}_{\text{LS}} - \boldsymbol{\theta}^0 = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{V}, \tag{12.91}$$

and the bias is

$$
\begin{aligned}
\mathcal{E}\hat{\boldsymbol{\theta}}_{\text{LS}} - \boldsymbol{\theta}^0 &= \mathcal{E}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{V}\\
&= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathcal{E}\mathbf{V}\\
&= \mathbf{0}.
\end{aligned}
\tag{12.92}
$$

Actually, we would really like to have $\hat{\boldsymbol{\theta}}(N)$ be "close" to $\boldsymbol{\theta}^0$ for finite values of N and would, as a third property, like to treat the mean-square parameter error for finite numbers of samples. Unfortunately, it is difficult to obtain an estimate that has a minimum for the square of $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0$ without involving the value of $\boldsymbol{\theta}^0$ directly, which we do not know, else there is no point in estimating it. We can, however, find an estimate that is the best (in the sense of mean-square parameter error) estimate which is also linear in $\mathbf{Y}$ and unbiased. The result is called a Best Linear Unbiased Estimate, or BLUE.

The development proceeds as follows. Because the estimate is to be a linear function of the data, we write[11]

$$\hat{\boldsymbol{\theta}} = \mathbf{L} \cdot \mathbf{Y}. \tag{12.93}$$

Because the estimate is to be unbiased, we require

$$\mathcal{E}\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}^0,$$

or

$$\mathcal{E}\mathbf{LY} = \mathcal{E}\mathbf{L}(\mathbf{A}\boldsymbol{\theta}^0 + \mathbf{V}) = \boldsymbol{\theta}^0.$$

Thus

$$\mathbf{LA}\boldsymbol{\theta}^0 = \boldsymbol{\theta}^0,$$

or

$$\mathbf{LA} = \mathbf{I}. \tag{12.94}$$

We wish to find $\mathbf{L}$ so that an estimate of the form Eq. (12.93), subject to the constraint Eq. (12.94), makes the mean-square error

$$\mathcal{J}(\mathbf{L}) = \operatorname{tr}\mathcal{E}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)^T \tag{12.95}$$

as small as possible. Using Eq. (12.93) in Eq. (12.95), we have

$$\mathcal{J}(\mathbf{L}) = \operatorname{tr}\mathcal{E}\{(\mathbf{LY} - \boldsymbol{\theta}^0)(\mathbf{LY} - \boldsymbol{\theta}^0)^T\},$$

and using Eq. (12.77) for $\mathbf{Y}$

$$\mathcal{J}(\mathbf{L}) = \operatorname{tr}\mathcal{E}(\mathbf{LA}\boldsymbol{\theta}^0 + \mathbf{LV} - \boldsymbol{\theta}^0)(\mathbf{LA}\boldsymbol{\theta}^0 + \mathbf{LV} - \boldsymbol{\theta}^0)^T.$$

But from Eq. (12.94), this reduces to

$$\begin{aligned}\mathcal{J}(\mathbf{L}) &= \operatorname{tr}\mathcal{E}(\mathbf{LV})(\mathbf{LV})^T \\ &= \operatorname{tr}\mathbf{LRL}^T,\end{aligned} \tag{12.96}$$

where we take the covariance for the noise to be

$$\mathcal{E}\mathbf{VV}^T = \mathbf{R}.$$

We now have the entirely deterministic problem of finding that $\mathbf{L}$ subject to Eq. (12.94) which makes Eq. (12.96) as small as possible. We solve the problem in an indirect way: We first find $\mathbf{L}$ when $\boldsymbol{\theta}^0$ is a scalar and there is no trace operation ($\mathbf{L}$ is a single row). From this scalar solution we *conjecture* what the multiparameter solution might be and demonstrate that it is in fact correct. First, we consider the case when $\mathbf{L}$ is a row. We must introduce the constraint

---

11 The matrix $\mathbf{L}$ used here for a linear dependence has no connection to the gain matrix used in the recursive estimate equation (12.72).

Eq. (12.94), and we do this by a Lagrange multiplier as in calculus and are led to find $\mathbf{L}$ such that

$$\mathcal{J}(\mathbf{L}) = \mathbf{LRL}^T + \lambda(\mathbf{A}^T\mathbf{L}^T - \mathbf{I}) \tag{12.97}$$

is a minimum. The λ are the Lagrange multipliers. The necessary conditions on the elements of $\mathbf{L}$ are that $\partial\mathcal{J}/\partial\mathbf{L}^T$ be zero. (We use $\mathbf{L}^T$, which is a column, to retain notation that is consistent with our earlier discussion of vector-matrix derivatives.) We have

$$\left.\frac{\partial\mathcal{J}}{\partial\mathbf{L}^T}\right|_{\mathbf{L}=\hat{\mathbf{L}}} = 2\hat{\mathbf{L}}\mathbf{R} + \lambda\mathbf{A}^T = 0.$$

Thus the best value for $\mathbf{L}$ is given by

$$\begin{aligned} 2\hat{\mathbf{L}}\mathbf{R} &= -\lambda\mathbf{A}^T, \\ \hat{\mathbf{L}} &= -\frac{1}{2}\lambda\mathbf{A}^T\mathbf{R}^{-1}. \end{aligned} \tag{12.98}$$

Because the constraint Eq. (12.94) must be satisfied, $\hat{\mathbf{L}}$ must be such that

$$\hat{\mathbf{L}}\mathbf{A} = \mathbf{I},$$

or

$$-\frac{1}{2}\lambda\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A} = \mathbf{I}.$$

From this we conclude that

$$\lambda = -2(\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1}, \tag{12.99}$$

and substituting Eq. (12.99) back again in Eq. (12.98), we have, finally, that

$$\hat{\mathbf{L}} = (\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1}\mathbf{A}^T\mathbf{R}^{-1}, \tag{12.100}$$

best linear unbiased estimate (BLUE)

and the BLUE is

$$\hat{\boldsymbol{\theta}}_B = \hat{\mathbf{L}}\mathbf{Y} = (\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1}\mathbf{A}^T\mathbf{R}^{-1}\mathbf{Y}. \tag{12.101}$$

If we look at Eq. (12.61), we immediately see that this is exactly weighted least squares with $\mathbf{W} = \mathbf{R}^{-1}$. What we have done, in effect, is give a reasonable—best linear unbiased—criterion for selection of the weights. If, as we assumed earlier, $\mathbf{R} = \sigma^2\mathbf{I}$, then the ordinary least-squares estimate is also the BLUE.

But we have jumped ahead of ourselves; we have yet to show that Eq. (12.100) or Eq. (12.101) is true for a matrix $\mathbf{L}$. Suppose we have another linear unbiased estimate, $\overline{\boldsymbol{\theta}}$. We can write, without loss of generality

$$\overline{\boldsymbol{\theta}} = \mathbf{LY} = (\hat{\mathbf{L}} + \overline{\mathbf{L}})\mathbf{Y},$$

where $\hat{\mathbf{L}}$ is given by Eq. (12.100). Because $\bar{\boldsymbol{\theta}}$ is required to be unbiased, Eq. (12.94) requires that

$$\begin{gathered}\mathbf{LA} = \mathbf{I}, \quad (\hat{\mathbf{L}} + \bar{\mathbf{L}})\mathbf{A} = \mathbf{I}, \quad \hat{\mathbf{L}}\mathbf{A} + \bar{\mathbf{L}}\mathbf{A} = \mathbf{I}, \\ \mathbf{I} + \bar{\mathbf{L}}\mathbf{A} = \mathbf{I}, \quad \bar{\mathbf{L}}\mathbf{A} = \bar{0}.\end{gathered} \tag{12.102}$$

Now the mean-square error using $\bar{\boldsymbol{\theta}}$ is

$$\mathcal{J}(\bar{\boldsymbol{\theta}}) = \operatorname{tr}\mathcal{E}(\bar{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)(\bar{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)^T,$$

which can be written as

$$\begin{aligned}\mathcal{J}(\bar{\boldsymbol{\theta}}) &= \operatorname{tr}\mathcal{E}(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}} + \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}} + \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)^T \\ &= \operatorname{tr}\{\mathcal{E}(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}})(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}})^T + 2\mathcal{E}(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)^T + \mathcal{E}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)^T\}.\end{aligned} \tag{12.103}$$

We note that the trace of a sum is the sum of the traces of the components, and the trace of the last term is, by definition, $\mathcal{J}(\hat{\mathbf{L}})$. Let us consider the second term of Eq. (12.103), namely

$$\begin{aligned}\text{term } 2 = \operatorname{tr}\mathcal{E}(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^0)^T &= \operatorname{tr}\mathcal{E}(\mathbf{LY} - \hat{\mathbf{L}}\mathbf{Y})(\hat{\mathbf{L}}\mathbf{Y} - \boldsymbol{\theta}^0)^T \\ &= \operatorname{tr}\mathcal{E}(\hat{\mathbf{L}}\mathbf{Y} + \bar{\mathbf{L}}\mathbf{Y} - \hat{\mathbf{L}}\mathbf{Y})(\hat{\mathbf{L}}\mathbf{Y} - \boldsymbol{\theta}^0) \\ &= \operatorname{tr}\mathcal{E}(\bar{\mathbf{L}}\mathbf{Y})(\hat{\mathbf{L}}\mathbf{Y} - \boldsymbol{\theta}^0) \\ &= \operatorname{tr}\mathcal{E}(\bar{\mathbf{L}}(\mathbf{A}\boldsymbol{\theta}^0 + \mathbf{V}))(\hat{\mathbf{L}}(\mathbf{A}\boldsymbol{\theta}^0 + \mathbf{V}) - \boldsymbol{\theta}^0)^T.\end{aligned}$$

Now we use Eq. (12.102) to eliminate one term and Eq. (12.94) to eliminate another to the effect that

$$\begin{aligned}\text{term } 2 &= \operatorname{tr}\mathcal{E}(\bar{\mathbf{L}}\mathbf{V})(\hat{\mathbf{L}}\mathbf{V})^T \\ &= \operatorname{tr}\bar{\mathbf{L}}\mathbf{R}\hat{\mathbf{L}}^T,\end{aligned}$$

and using Eq. (12.100) for $\hat{\mathbf{L}}$, we have

$$\begin{aligned}\text{term } 2 &= \operatorname{tr}\bar{\mathbf{L}}\mathbf{R}\{\mathbf{R}^{-1}\mathbf{A}(\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1}\} \\ &= \operatorname{tr}\bar{\mathbf{L}}\mathbf{A}(\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1},\end{aligned}$$

but now from Eq. (12.102) we see that the term is zero. Thus we reduce Eq. (12.103) to

$$\mathcal{J}(\bar{\boldsymbol{\theta}}) = \operatorname{tr}\mathcal{E}(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}})(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}})^T + \mathcal{J}(\hat{\boldsymbol{\theta}}).$$

Because the first term on the right is the sum of expected values of squares, it is zero or positive. Thus $\mathcal{J}(\bar{\boldsymbol{\theta}}) \geq \mathcal{J}(\hat{\boldsymbol{\theta}})$, and we have proved that Eq. (12.101) is really and truly BLUE.

To conclude this section on stochastic least squares, we summarize our findings as follows. If the data are described by

$$\mathbf{Y} = \mathbf{A}\boldsymbol{\theta}^0 + \mathbf{V},$$

and

$$\mathcal{E}\mathbf{V} = 0, \qquad \mathcal{E}\mathbf{V}\mathbf{V}^T = \mathbf{R},$$

then the least-squares estimate of $\boldsymbol{\theta}^0$ is given by

$$\hat{\boldsymbol{\theta}}_{LS} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{Y}, \tag{12.104}$$

which is unbiased. If $\mathbf{R} = \sigma^2\mathbf{I}$, then the variance of $\hat{\boldsymbol{\theta}}_{LS}$, which is defined as $\mathcal{E}(\hat{\boldsymbol{\theta}}_{LS} - \boldsymbol{\theta}^0)(\hat{\boldsymbol{\theta}}_{LS} - \boldsymbol{\theta}^0)^T$, is

$$\text{var}(\hat{\boldsymbol{\theta}}_{LS}) = \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}. \tag{12.105}$$

From this we showed that the least-squares estimation of a constant in noise is not only unbiased but also consistent.

Furthermore, if we insist that the estimate be both a linear function of the data and unbiased, we showed that the BLUE is a weighted least squares given by

$$\hat{\boldsymbol{\theta}}_B = (\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1}\mathbf{A}^T\mathbf{R}^{-1}\mathbf{Y}. \tag{12.106}$$

The variance of $\hat{\boldsymbol{\theta}}_B$ is

$$\begin{aligned} \text{var}(\hat{\boldsymbol{\theta}}_B) &= \mathcal{E}(\hat{\boldsymbol{\theta}}_B - \boldsymbol{\theta}^0)(\hat{\boldsymbol{\theta}}_B - \boldsymbol{\theta}^0)^T \\ &= (\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1}. \end{aligned} \tag{12.107}$$

Thus, if $\mathbf{R} = \sigma^2\mathbf{I}$, then the least-squares estimate is also the BLUE.

As another comment on the BLUE, we note that in a recursive formulation we take, according to Eq. (12.63)

$$\mathbf{P} = (\mathbf{A}^T\mathbf{R}^{-1}\mathbf{A})^{-1},$$

and so the matrix $\mathbf{P}$ is the variance of the estimate $\hat{\boldsymbol{\theta}}_B$. In the recursive equations given for vector measurements in Eq. (12.75), the weight matrix a becomes $\mathbf{R}_v^{-1}$, the inverse of the covariance of the single time measurement noise vector and, of course, $\gamma = 1$. Thus the BLUE version of Eq. (12.75) is

$$\mathbf{L}(N+1) = \mathbf{P}\boldsymbol{\phi}(\mathbf{R}_v + \boldsymbol{\phi}^T\mathbf{P}\boldsymbol{\phi})^{-1}.$$

Thus far we have considered only the least-squares estimation where $\mathbf{A}$, the coefficient of the unknown parameter, is known. However, as we have seen earlier, in the true identification problem, the coefficient of $\boldsymbol{\theta}$ is $\boldsymbol{\phi}$, the state of the plant model; and if random noise is present, the elements of $\boldsymbol{\phi}$ will be random variables also. Analysis in this case is complex, and the best we will be able to do is to quote results on consistency and bias before turning to the method of maximum likelihood. We can, however, illustrate the major features and the major difficulty of least-squares identification by analysis of a very simple case.

Suppose we consider a first-order model with no control. The equations are taken to be

$$\begin{aligned} y(k) &= a^0 y(k-1) + v(k) + cv(k-1), \\ \mathcal{E}v(k) &= 0, \\ \mathcal{E}v(k)v(j) &= \begin{cases} \sigma^2 & k = j \\ 0 & k \neq j. \end{cases} \end{aligned} \tag{12.108}$$

In general, we would not know either the constant c or the noise intensity σ^2. We will estimate them later. For the moment, we wish to consider the effects of the noise on the least-squares estimation of the constant a^0. Thus we take

$$\begin{aligned} \mathbf{Y} &= [y(1) \cdots y(N)]^T, \\ \boldsymbol{\phi} &= [y(0) \cdots y(N-1)]^T, \\ \theta &= a. \end{aligned}$$

Then we have the simple sums

$$\boldsymbol{\phi}^T\boldsymbol{\phi} = \sum_{k=1}^{N} y(k-1)y(k-1), \qquad \boldsymbol{\phi}^T\mathbf{Y} = \sum_{k=1}^{N} y(k-1)y(k),$$

and the normal equations tell us that $\hat{\boldsymbol{\theta}}$ must satisfy

$$\left[\sum_{k=1}^{N} y^2(k-1)\right]\hat{\theta} = \sum_{k=1}^{N} y(k-1)y(k). \tag{12.109}$$

Now we must quote a result from statistics (see Appendix D). For signals such as $y(k)$ that are generated by white noise of zero mean having finite intensity passed through a stationary filter, we can define the autocorrelation function

$$R_y(j) = \mathcal{E}y(k)y(k+j) \tag{12.110}$$

and (in a suitable sense of convergence)

$$\lim_{N\to\infty} \frac{1}{N}\sum_{k=1}^{N} y(k)y(k+j) = R_y(j). \tag{12.111}$$

Thus, while the properties of $\hat{\theta}$ for finite N are difficult, we can say that the asymptotic least-squares estimate is given by the solution to

$$\lim_{N\to\infty}\left(\frac{1}{N}\sum_{k=1}^{N} y^2(k-1)\right)\hat{\theta} = \lim_{N\to\infty}\frac{1}{N}\sum_{k=1}^{N} y(k-1)y(k),$$

which is

$$\begin{aligned} R_y(0)\hat{\theta} &= R_y(1), \\ \hat{\theta} &= R_y(1)/R_y(0). \end{aligned} \tag{12.112}$$

If we now return to the model from which we assume $y(k)$ to be generated, values for $R_y(0)$ and $R_y(1)$ can be obtained. For example, if we multiply Eq. (12.108) by $y(k-1)$ and take the expected value, we find

$$\begin{aligned} \mathcal{E}y(k-1)y(k) &= \mathcal{E}a^0 y(k-1)y(k-1) + \mathcal{E}v(k)y(k-1) \\ &\quad + c\mathcal{E}v(k-1)y(k-1), \\ R_y(1) &= a^0 R_y(0) + c\mathcal{E}v(k-1)y(k-1). \end{aligned} \tag{12.113}$$

We set $\mathcal{E}v(k)y(k-1) = 0$ because $y(k-1)$ is generated by $v(j)$ occurring at or before time $k-1$ and y is independent of (uncorrelated with) the noise $v(j)$, which is in the future. However, we still need to compute $\mathcal{E}v(k-1)y(k-1)$. For this, we write Eq. (12.108) for $k-1$ as

$$y(k-1) = a^0 y(k-2) + v(k-1) + cv(k-2)$$

and multiply by $v(k-1)$ and take expected values,

$$\begin{aligned} \mathcal{E}y(k-1)v(k-1) &= \mathcal{E}a^0 y(k-2)v(k-1) + \mathcal{E}v^2(k-1) + c\mathcal{E}v(k-2)v(k-1) \\ &= \sigma^2. \end{aligned}$$

We have, then,

$$R_y(1) = a^0 R_y(0) + c\sigma^2. \tag{12.114}$$

Now we can substitute these values into Eq. (12.112) to obtain

$$\begin{aligned} \hat{\theta}(\infty) &= \frac{R_y(1)}{R_y(0)} \\ &= a^0 + c\frac{\sigma^2}{R_y(0)}. \end{aligned} \tag{12.115}$$

If $c = 0$, then $\hat{\theta}(\infty) = a^0$, and we can say that the least-squares estimate is asymptotically unbiased. However, if $c \neq 0$, Eq. (12.115) shows that even in this simple case the least-squares estimate is asymptotically biased and cannot be consistent.[12] Primarily because of the bias introduced when the ARMA model has noise terms correlated from one equation to the next, as in Eq. (12.108) when $c \neq 0$, least squares is not a good scheme for constructing parameter estimates of dynamic models that include noise. Many alternatives have been studied, among the most successful being those based on maximum likelihood.

12 Clearly, if the constant c is known we can subtract out the bias term, but such corrections are not very pleasing because the essential nature of noise is its unknown dependencies. In some instances it is possible to construct a term that asymptotically cancels the bias term as discussed in Mendel (1973).

12.6 Maximum Likelihood

The method of maximum likelihood requires that we introduce a probability density function for the random variables involved. Here we consider only the normal or Gaussian distribution; the method is not restricted to the form of the density function in any way, however. A scalar random variable x is said to have a normal distribution if it has a density function given by

$$f_x(\xi) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\frac{(\xi-\mu)^2}{\sigma^2}\right]. \tag{12.116}$$

It can be readily verified, perhaps by use of a table of definite integrals, that

$$\begin{aligned}
&\int_{-\infty}^{\infty} f_x(\xi)d\xi = 1, \\
\mathcal{E}x &= \int_{-\infty}^{\infty} \xi f_x(\xi)d\xi = \mu, \\
\mathcal{E}(x-\mu)^2 &= \int_{-\infty}^{\infty} (\xi-\mu)^2 f_x(\xi)d\xi = \sigma^2.
\end{aligned} \tag{12.117}$$

The number σ^2 is called the variance of x, written var(x), and σ is the standard deviation. For the case of a vector-valued set of n random variables with a joint distribution that is normal, we find that if we define the mean vector $\boldsymbol{\mu}$ and the (nonsingular) covariance matrix $\mathbf{R}$ as

$$\begin{aligned}
\mathcal{E}\mathbf{x} &= \boldsymbol{\mu}, \\
\mathcal{E}(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^T &= \mathbf{R},
\end{aligned}$$

then

$$f_x(\xi) = \frac{1}{[(2\pi)^n \det \mathbf{R}]^{1/2}} \exp\left[-\frac{1}{2}(\boldsymbol{\xi}-\boldsymbol{\mu})^T\mathbf{R}^{-1}(\boldsymbol{\xi}-\boldsymbol{\mu})\right]. \tag{12.118}$$

If the elements of the vector $\mathbf{x}$ are mutually uncorrelated and have identical means μ and variances σ^2, then $\mathbf{R} = \sigma^2\mathbf{I}$, $det\mathbf{R} = (\sigma^2)^n$, and Eq. (12.118) can be written as

$$f_x(\xi) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(\xi_i-\mu)^2\right]. \tag{12.119}$$

Because the normal distribution is completely determined by the vector of means $\boldsymbol{\mu}$ and the covariance matrix $\mathbf{R}$, we often use the notation $\mathcal{N}(\boldsymbol{\mu}, \mathbf{R})$ to designate the density Eq. (12.118).

The maximum-likelihood estimate is calculated on the basis of an assumed structure for the probability density function of the available observations. Suppose, for example, that the data consist of a set of observations having a density given by Eq. (12.119), but having an *unknown* mean value. The parameter is therefore $\theta^0 = \mu$.

The function[13]

$$f_x(\boldsymbol{\xi} \mid \theta) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2} \sum_{i=1}^{n} \frac{(\xi_i - \theta)^2}{\sigma^2} \right] \tag{12.120}$$

can be presented *as a function of* θ, as giving the density of a set of x_i for *any* value of the population mean θ. Because the probability that a particular x_i is in the range $a \le x_i \le b$ is given by

$$\Pr\{a \le x_i \le b\} = \int_a^b f_{x_i}(\xi_i \mid \theta) d\xi_i,$$

the density function is seen to be a measure of the "likelihood" for a particular value; when f is large in a neighborhood $\mathbf{x}^0$, we would expect to find many samples from the population with values near $\mathbf{x}^0$. As a function of the parameters, $\boldsymbol{\theta}$, the density function $f_x(\boldsymbol{\xi} \mid \boldsymbol{\theta})$ is called the *likelihood function*. If the actual data come from a population with the density $f_x(\boldsymbol{\xi} \mid \boldsymbol{\theta}^0)$, then one might expect the samples to reflect this fact and that a good estimate for $\boldsymbol{\theta}^0$ given the observations $\mathbf{x} = \{x_1, \ldots, x_n\}^T$ would be $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$, where $\hat{\boldsymbol{\theta}}$ is such that the likelihood function $f_x(\mathbf{x} \mid \boldsymbol{\theta})$ is as large as possible. Such an estimate is called the maximum-likelihood estimate, $\hat{\boldsymbol{\theta}}_{\mathrm{ML}}$. Formally, $\hat{\boldsymbol{\theta}}_{\mathrm{ML}}$ is such that

$$f_x(\mathbf{x} \mid \hat{\boldsymbol{\theta}}_{\mathrm{ML}}) \ge f_x(\mathbf{x} \mid \boldsymbol{\theta}). \tag{12.121}$$

From Eq. (12.120) we can immediately compute $\hat{\theta}_{\mathrm{ML}}$ for the mean μ, by setting the derivative of $f_x(\mathbf{x} \mid \theta)$ with respect to θ equal to zero. First, we note that

$$\frac{d}{d\theta} \log f = \frac{1}{f} \frac{df}{d\theta}, \tag{12.122}$$

so that the derivative of the log of f is zero when $df/d\theta$ is zero.[14] Because the (natural) log of the normal density is a simpler function than the density itself, we will often deal with the log of f. In fact, the negative of the log of f is used so much, we will call it the log-likelihood function and give it the functional designation

$$-\log f_x(\mathbf{x} \mid \boldsymbol{\theta}) = \ell(\mathbf{x} \mid \boldsymbol{\theta}).$$

Thus, from Eq. (12.120), for the scalar parameter μ, we have

$$\ell(\mathbf{x} \mid \theta) = +\frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2} \sum_{i=1}^{n} \frac{(x_i - \theta)^2}{\sigma^2} \tag{12.123}$$

13 We read $f_x(\boldsymbol{\xi} \mid \theta)$ as "the probability density of $\boldsymbol{\xi}$ given θ."

14 It is not possible for f to be zero in the neighborhood of its maximum.

and

$$\frac{\partial \ell}{\partial \theta} = -\frac{1}{2\sigma^2}\sum_{i=1}^{n} 2(x_i - \theta)$$

$$= -\frac{1}{\sigma^2}\left\{\sum_{i=1}^{n} x_i - n\theta\right\}. \tag{12.124}$$

If we now set $\partial \ell / \partial \theta = 0$, we have

$$\sum_{i=1}^{n} x_i - n\hat{\theta}_{\mathrm{ML}} = 0$$

or

$$\hat{\theta}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{12.125}$$

We thus find that for an unknown mean of a normal distribution, the maximum-likelihood estimate is the sample mean, which is also least squares, unbiased, consistent, and BLUE. We have studied this estimate before. However, we can go on to apply the principle of maximum likelihood to the problem of dynamic system identification.

Consider next the ARMA model with simple white-noise disturbances for which we write

$$\begin{aligned} y(k) = &-a_1 y(k-1) - \cdots - a_n y(k-n) \\ &+ b_1 u(k-1) + \cdots + b_n u(k-n) + v(k), \end{aligned} \tag{12.126}$$

and we assume that the distribution of $\mathbf{V} = [v(n) \ldots v(N)]^T$ is $N(0, \sigma^2\mathbf{I})$. Thus we are assuming that the $v(k)$ are each normally distributed with zero mean and variance σ^2 and furthermore that the covariance between $v(k)$ and $v(j)$ is zero for $k \neq j$. Suppose, now, that a sequence of the $y(k)$ are observed and that we wish to estimate from them the a_i, b_i, and σ^2 by the method of maximum likelihood. We require the probability-density function of the observed $y(k)$ for known values of the parameters. From a look at Eq. (12.126), it is apparent that if we assume that y, u, a_i, and b_i are all known, then we can compute $v(k)$ from these observed y and u and assumed (true) a_i and b_i, and the distribution of y is immediately determined by the distribution of v. Using the earlier notation, we define

$$\begin{aligned} \mathbf{V}(N) &= [v(n) \ldots v(N)]^T, \\ \mathbf{Y}(N) &= [y(n) \ldots y(N)]^T, \\ \boldsymbol{\phi}(k) &= [-y(k-1) \ldots - y(k-n) u(k-1) \ldots u(k-n)]^T, \\ \boldsymbol{\Phi}(N) &= [\boldsymbol{\phi}(n) \ldots \boldsymbol{\phi}(N)]^T, \\ \boldsymbol{\theta}^0 &= [a_1 \ldots a_n b_1 \ldots b_n]^T. \end{aligned} \tag{12.127}$$

Then Eq. (12.126) implies, again for the true parameters, that

$$\mathbf{Y}(N) = \mathbf{\Phi}\boldsymbol{\theta}^0 + \mathbf{V}(N). \tag{12.128}$$

Equation (12.128) is an expression of the input–output relation of our plant equations of motion. To obtain the probability-density function $f(\mathbf{Y} \mid \boldsymbol{\theta}^0)$ as required for the method of maximum likelihood, we need only be able to compute the $v(k)$ or in batch form, the $\mathbf{V}(N)$, from the $y(k)$ or $\mathbf{Y}(N)$ because we are given the probability density of $\mathbf{V}$. To compute $\mathbf{V}$ from $\mathbf{Y}$ requires the *inverse* model of our plant, which in this case is so trivial as to be almost missed;[15] namely, we solve Eq. (12.128) for $\mathbf{V}$ to obtain

$$\mathbf{V}(N) = \mathbf{Y}(N) - \mathbf{\Phi}(N)\boldsymbol{\theta}^0. \tag{12.129}$$

Because we have assumed that the density function of $\mathbf{V}$ is $\mathcal{N}(0, \sigma^2\mathbf{I})$, we can write instantly

$$f(\mathbf{Y} \mid \boldsymbol{\theta}^0) = (2\pi\sigma^2)^{-m/2} \exp\left[-\frac{1}{2}\frac{(\mathbf{Y} - \mathbf{\Phi}\boldsymbol{\theta}^0)^T(\mathbf{Y} - \mathbf{\Phi}\boldsymbol{\theta}^0)}{\sigma^2}\right], \tag{12.130}$$

where $m = N - n + 1$, the number of samples in Y.

The likelihood function is by definition $f(\mathbf{Y} \mid \boldsymbol{\theta})$, which is to say, Eq. (12.130) with the $\boldsymbol{\theta}^0$ dropped and replaced by a general $\boldsymbol{\theta}$. As in the elementary example discussed above, we will consider the negative of the log of f as follows

$$\begin{aligned}\ell(\mathbf{Y} \mid \boldsymbol{\theta}) &= -\log(2\pi\sigma^2)^{-m/2} - \log\left\{\exp\left[-\frac{1}{2}\frac{(\mathbf{Y} - \mathbf{\Phi}\boldsymbol{\theta})^T(\mathbf{Y} - \mathbf{\Phi}\boldsymbol{\theta})}{\sigma^2}\right]\right\} \\ &= +\frac{m}{2}\log 2\pi + \frac{m}{2}\log\sigma^2 + \frac{1}{2}\frac{(\mathbf{Y} - \mathbf{\Phi}\boldsymbol{\theta})^T(\mathbf{Y} - \mathbf{\Phi}\boldsymbol{\theta})}{\sigma^2}.\end{aligned} \tag{12.131}$$

Our estimates, $\hat{\boldsymbol{\theta}}_{\text{ML}}$ and $\hat{\sigma}^2_{\text{ML}}$, are those values of $\boldsymbol{\theta}$ and σ^2 which make $\ell(\mathbf{Y} \mid \boldsymbol{\theta})$ as small as possible. We find these estimates by setting to zero the partial derivatives of ℓ with respect to $\boldsymbol{\theta}$ and σ^2. These derivatives are (following our earlier treatment of taking partial derivatives)

$$\begin{aligned}\frac{\partial \ell}{\partial \boldsymbol{\theta}} &= \frac{1}{\hat{\sigma}^2}[\mathbf{\Phi}^T\mathbf{\Phi}\hat{\boldsymbol{\theta}}_{\text{ML}} - \mathbf{\Phi}^T\mathbf{Y}] = \mathbf{0}, && (a) \\ \frac{\partial \ell}{\partial \sigma^2} &= \frac{m}{2\hat{\sigma}^2_{\text{ML}}} - \frac{\mathbf{Q}}{2\hat{\sigma}^4_{\text{ML}}} = 0, && (b)\end{aligned} \tag{12.132}$$

where the quadratic term $\mathbf{Q}$ is defined as

$$\mathbf{Q} = (\mathbf{Y} - \mathbf{\Phi}\hat{\boldsymbol{\theta}}_{\text{ML}})^T(\mathbf{Y} - \mathbf{\Phi}\hat{\boldsymbol{\theta}}_{\text{ML}}). \qquad (c)$$

15 We deliberately formulated our problem so that this inverse would be trivial. If we add plant and independent sensor noise to a state-variable description, the inverse requires a Kalman filter. Because $v(k)$ is independent of all past u and y for any $\boldsymbol{\theta}$, $e(k)$ is the prediction error.

We see immediately that Eq. (12.132) is the identical "normal" equation of the least-squares method so that $\hat{\boldsymbol{\theta}}_{\text{ML}} = \hat{\boldsymbol{\theta}}_{\text{LS}}$ in this case. We thus know that $\hat{\boldsymbol{\theta}}_{\text{ML}}$ is asymptotically unbiased and consistent. The equations for $\hat{\sigma}^2_{\text{ML}}$ decouple from those for $\hat{\boldsymbol{\theta}}_{\text{ML}}$, and solving Eq. (12.132(b)), we obtain (again using earlier notation)

$$\begin{aligned}\hat{\sigma}^2_{\text{ML}} &= \frac{\mathbf{Q}}{m} = \frac{1}{m}(\mathbf{Y} - \boldsymbol{\Phi}\hat{\boldsymbol{\theta}})^T(\mathbf{Y} - \boldsymbol{\Phi}\hat{\boldsymbol{\theta}}) \\ &= \frac{1}{N-n+1}\sum_{k=n}^{N} e^2(k;\, \hat{\boldsymbol{\theta}}_{\text{ML}}). \qquad (12.133)\end{aligned}$$

Thus far we have no new solutions except the estimate for σ^2 given in Eq. (12.133), but we have shown that the method of maximum likelihood gives the same solution for $\hat{\boldsymbol{\theta}}$ as the least-squares method for the model of Eq. (12.126). Now let us consider the general model given by

$$y(k) = -\sum_{i=1}^{n} a_i y(k-i) + \sum_{i=1}^{n} b_i u(k-i) + \sum_{i=1}^{n} c_i v(k-i) + v(k); \quad (12.134)$$

the distribution of $\mathbf{V}(N) = [v(n) \ldots v(N)]^T$ is again taken to be normal, and $\mathbf{V}$ is distributed according to the $N(0, \sigma^2\mathbf{I})$ density. The difference between Eq. (12.134) and Eq. (12.126), of course, is that in Eq. (12.134) we find past values of the noise $v(k)$ weighted by the c_i and, as we saw in Eq. (12.115), the least-squares estimate is biased if the c_i are nonzero.

Consider first the special case where of the c's only c_1 is nonzero, and write the terms in Eq. (12.134) that depend on noise $v(k)$ on one side and define $z(k)$ as follows

$$\begin{aligned}v(k) + c_1 v(k-1) &= y(k) + \sum_{i=1}^{n} a_i y(k-i) - \sum_{i=1}^{n} b_i u(k-i) \\ &= z(k). \qquad (12.135)\end{aligned}$$

Now let the reduced parameter vector be $\bar{\boldsymbol{\theta}} = [a_1 \ldots a_n \quad b_1 \ldots b_n]^T$, composed of the a and b parameters but not including the c_i. In a natural way, we can write

$$\mathbf{Z}(N) = \mathbf{Y}(N) - \boldsymbol{\Phi}\bar{\boldsymbol{\theta}}^0. \quad (12.136)$$

However, because $z(k)$ is a sum of two random variables $v(k)$ and $v(k-1)$, each of which has a normal distribution, we know that $\mathbf{Z}$ is also normally distributed. Furthermore, we can easily compute the mean and covariance of z

$$\begin{aligned}\mathcal{E}z(k) &= \mathcal{E}(v(k) + c_1 v(k-1)) = 0 \quad \text{for all } k, \\ \mathcal{E}z(k)z(j) &= \mathcal{E}(v(k) + c_1 v(k-1))(v(j) + c_1 v(j-1)) \\ &= \sigma^2(1 + c_1^2) \qquad (k = j) \\ &= \sigma^2 c_1 \qquad (k = j-1) \\ &= \sigma^2 c_1 \qquad (k = j+1) \\ &= 0 \qquad \text{elsewhere.}\end{aligned}$$

Thus the structure of the covariance of $\mathbf{Z}(N)$ is

$$\mathbf{R} = \mathcal{E}\mathbf{Z}(N)\mathbf{Z}^T(N)$$
$$= \begin{bmatrix} 1+c_1^2 & c_1 & 0 & 0 & 0 \\ c_1 & 1+c_1^2 & c_1 & 0 & 0 \\ 0 & c_1 & 1+c_1^2 & \ddots & \vdots \\ 0 & 0 & \ddots & \ddots & c_1 \\ 0 & 0 & \cdots & c_1 & 1+c_1^2 \end{bmatrix} \sigma^2, \tag{12.137}$$

and, with the mean and covariance in hand, we can write the probability density of $\mathbf{Z}(N)$ as

$$g(\mathbf{Z}(N) \mid \boldsymbol{\theta}^0) = ((2\pi)^m \det \mathbf{R})^{-1/2} \exp(-\frac{1}{2}\mathbf{Z}^T\mathbf{R}^{-1}\mathbf{Z}), \tag{12.138}$$

where $m = N - n + 1$. But from (12.136) this is the likelihood function if we substitute for $\mathbf{Z}$ and $\boldsymbol{\theta}$ as follows:

$$f(\mathbf{Y} \mid \boldsymbol{\theta}) = [(2\pi)^m \det \mathbf{R}]^{-1/2} \exp[-\frac{1}{2}(\mathbf{Y} - \boldsymbol{\Phi}\bar{\boldsymbol{\theta}})^T\mathbf{R}^{-1}(\mathbf{Y} - \boldsymbol{\Phi}\bar{\boldsymbol{\theta}})]. \tag{12.139}$$

The negative of the log of f is again similar in form to previous results

$$\ell(\mathbf{Y} \mid \boldsymbol{\theta}) = \frac{m}{2}\log 2\pi + \frac{1}{2}\log(\det \mathbf{R}) + \frac{1}{2}(\mathbf{Y} - \boldsymbol{\Phi}\bar{\boldsymbol{\theta}})^T\mathbf{R}^{-1}(\mathbf{Y} - \boldsymbol{\Phi}\bar{\boldsymbol{\theta}}). \tag{12.140}$$

The major point to be made about Eq. (12.140) is that the log likelihood function depends on the a's and b's through $\bar{\boldsymbol{\theta}}$ and is thus *quadratic* in these parameters, but it depends on the c's (c_1 only in this special case) through both $\mathbf{R}$ and $det\mathbf{R}$, a dependence that is definitely not quadratic. An explicit formula for the maximum-likelihood estimate is thus not possible, and we must retreat to a numerical algorithm to compute $\hat{\boldsymbol{\theta}}_{ML}$ in this case.

12.7 Numerical Search for the Maximum-Likelihood Estimate

Being unable to give a closed-form expression for the maximum-likelihood estimate, we turn to an algorithm that can be used for the numerical search for $\hat{\boldsymbol{\theta}}_{ML}$. We first formulate the problem from the assumed ARMA model of Eq. (12.134), once again forming the inverse system as

$$v(k) = y(k) + \sum_{i=1}^{n} a_i y(k-i) - \sum_{i=1}^{n} b_i u(k-i) - \sum_{i=1}^{n} c_i v(k-i). \tag{12.141}$$

By assumption, $v(k)$ has a normal distribution with zero mean and unknown (scalar) variance R_v. As before, we define the successive outputs of this inverse as

$$\mathbf{V}(N) = [v(n) \quad v(n+1) \quad \cdots \quad v(N)]^T. \tag{12.142}$$

This multivariable random vector also has a normal distribution with zero mean and covariance

$$\begin{aligned} \mathbf{R} &= R_v \mathbf{I}_m \\ &= \mathcal{E}\mathbf{V}\mathbf{V}^T, \end{aligned} \tag{12.143}$$

where $\mathbf{I}_m$ is the $m \times m$ identity matrix, and $m = N - n + 1$ is the number of elements in $\mathbf{V}$. Thus we can immediately write the density function as a function of the true parameters $\boldsymbol{\theta}^0 = [a_1\, a_2\, \ldots\, a_n\, b_1\, \ldots\, b_n\, c_1\, \ldots\, c_n]^T$ as

$$f(\mathbf{V}|\boldsymbol{\theta}) = ((2\pi)^m R_v^m)^{-1/2} exp\left[-\frac{1}{2}\sum_{k=n}^{N}\frac{v^2(k)}{R_v}\right]. \tag{12.144}$$

The (log) likelihood function is found by substituting arbitrary parameters, $\boldsymbol{\theta}$, in Eq. (12.144) and using $e(k)$ as the output of Eq. (12.141) when $\boldsymbol{\theta} \neq \boldsymbol{\theta}^0$, then taking the log which gives the result

$$\ell(\mathbf{E}|\boldsymbol{\theta}) = \frac{m}{2} log 2\pi + \frac{m}{2} log \hat{R}_v + \frac{1}{2\hat{R}_v}\sum_{k=n}^{N} e^2(k). \tag{12.145}$$

As with Eq. (12.134) we can compute the estimate of $\hat{R}_v$ by calculation of

$$\partial\ell/\partial\hat{R}_v = 0,$$

which gives

$$\hat{R}_v = \frac{1}{N-n+1}\sum_{k=n}^{N} e^2(k). \tag{12.146}$$

To compute the values for the a_i, b_i, and c_i, we need to construct a numerical algorithm that will be suitable for minimizing $\ell(\mathbf{E}|\boldsymbol{\theta})$. The study of such algorithms is extensive;[16] we will be content to present a frequently used one, based on a method of Newton. The essential concept is that given the Kth estimate of $\hat{\boldsymbol{\theta}}$, we wish to find a $(K+1)$st estimate that will make $\ell(\mathbf{E}|\hat{\boldsymbol{\theta}}(K+1))$ smaller than $\ell(\mathbf{E}|\boldsymbol{\theta}(K))$. The method is to expand ℓ about $\hat{\boldsymbol{\theta}}(K)$ and choose $\hat{\boldsymbol{\theta}}(K+1)$ so that the quadratic terms—the first three terms in the expansion of ℓ—are minimized. Formally, we proceed as follows: Let $\hat{\boldsymbol{\theta}}(K+1) = \hat{\boldsymbol{\theta}}(K) + \delta\hat{\boldsymbol{\theta}}$. Then

$$\begin{aligned} \ell(\mathbf{E} \mid \hat{\boldsymbol{\theta}}(K+1)) &= \ell(\mathbf{E} \mid \hat{\boldsymbol{\theta}}(K) + \delta\hat{\boldsymbol{\theta}}) \\ &= c + \mathbf{g}^T\delta\hat{\boldsymbol{\theta}} + \frac{1}{2}\delta\hat{\boldsymbol{\theta}}^T\mathbf{Q}\delta\hat{\boldsymbol{\theta}} + \cdots, \end{aligned} \tag{12.147}$$

where

$$c = \ell(\mathbf{E} \mid \hat{\boldsymbol{\theta}}(K)), \quad \mathbf{g}^T = \left.\frac{\partial\ell}{\partial\boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(K)}, \quad \mathbf{Q} = \left.\frac{\partial^2\ell}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(K)}. \tag{12.148}$$

16 See Luenberger (1973) for a lucid account of minimizing algorithms.

We must return later to the computation of $\mathbf{g}^T$ and $\mathbf{Q}$, but let us first construct the algorithm. We would like to select $\delta\boldsymbol{\theta}$ so that the quadratic approximation to ℓ is as small as possible. The analytic condition for ℓ to be a minimum is that $\partial\ell/\partial\delta\boldsymbol{\theta} = 0$. We thus differentiate Eq. (12.147) and set the derivative to zero with the result (ignoring the higher-order terms in $\delta\boldsymbol{\theta}$)

$$\begin{aligned}\frac{\partial\ell}{\partial\delta\boldsymbol{\theta}} &= \mathbf{g}^T + \delta\boldsymbol{\theta}^T\mathbf{Q}\\ &= 0.\end{aligned} \tag{12.149}$$

Solving Eq. (12.149) for $\delta\boldsymbol{\theta}$ we find

$$\delta\boldsymbol{\theta} = -\mathbf{Q}^{-1}\mathbf{g}. \tag{12.150}$$

We now use the $\delta\boldsymbol{\theta}$ found in Eq. (12.150) to compute $\hat{\boldsymbol{\theta}}(K+1)$ as

$$\hat{\boldsymbol{\theta}}(K+1) = \hat{\boldsymbol{\theta}}(K) - \mathbf{Q}^{-1}\mathbf{g}.$$

In terms of ℓ as given in Eq. (12.148), the algorithm can be written as

$$\hat{\boldsymbol{\theta}}(K+1) = \hat{\boldsymbol{\theta}}(K) - \left(\frac{\partial^2\ell}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}}\right)^{-1}\left(\frac{\partial\ell}{\partial\boldsymbol{\theta}}\right)^T. \tag{12.151}$$

Our final task is to express the partial derivatives in Eq. (12.151) in terms of the observed signals y and u. To do this, we return to Eq. (12.145) and proceed formally, as follows, taking $\hat{R}_v$ to be a constant because $\hat{R}_v$ is given by Eq. (12.146)

$$\begin{aligned}\frac{\partial\ell(\mathbf{E}\,|\,\boldsymbol{\theta})}{\partial\boldsymbol{\theta}} &= \frac{\partial}{\partial\boldsymbol{\theta}}\left\{\frac{m}{2}\log 2\pi + \frac{m}{2}\log\hat{\mathbf{R}}_v + \frac{1}{2\hat{\mathbf{R}}_v}\sum_{k=n}^{N}e^2(k)\right\}\\ &= \frac{1}{\hat{\mathbf{R}}_v}\sum_{k=n}^{N}e(k)\frac{\partial e(k)}{\partial\boldsymbol{\theta}},\end{aligned} \tag{12.152}$$

$$\begin{aligned}\frac{\partial^2\ell}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}} &= \frac{\partial}{\partial\boldsymbol{\theta}}\left(\frac{1}{\hat{\mathbf{R}}_v}\sum_{k=n}^{N}e(k)\frac{\partial e(k)}{\partial\boldsymbol{\theta}}\right)\\ &= \frac{1}{\hat{\mathbf{R}}_v}\sum_{k=n}^{N}\left(\frac{\partial e(k)}{\partial\boldsymbol{\theta}}\right)^T\frac{\partial e(k)}{\partial\boldsymbol{\theta}} + \frac{1}{\hat{\mathbf{R}}_v}\sum_{k=n}^{N}e(k)\frac{\partial^2 e(k)}{\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}}.\end{aligned} \tag{12.153}$$

We note that Eq. (12.152) and the first term in Eq. (12.153) depend only on the derivative of e with respect to $\boldsymbol{\theta}$. Because our algorithm is expected to produce an improvement only in $\hat{\boldsymbol{\theta}}$, and because, near the minimum, we would expect the first derivative terms in Eq. (12.153) to dominate, we will simplify the algorithm to include only the first term in Eq. (12.153). Thus we need only compute $\partial e(k)/\partial\boldsymbol{\theta}$. It is standard terminology to refer to these partial derivatives as the *sensitivities* of e with respect to $\boldsymbol{\theta}$.

We turn to the difference equations for $v(k)$ Eq. (12.139), substitute $e(k)$ for $v(k)$, and compute the sensitivities as follows:

$$\frac{\partial e(k)}{\partial a_i} = y(k-i) - \sum_{j=1}^{n} c_j \frac{\partial e(k-j)}{\partial a_i} \qquad (a)$$

$$\frac{\partial e(k)}{\partial b_i} = -u(k-i) - \sum_{j=1}^{n} c_j \frac{\partial e(k-j)}{\partial b_i}, \qquad (b)$$

$$\frac{\partial e(k)}{\partial c_i} = -e(k-i) - \sum_{j=1}^{n} c_j \frac{\partial e(k-j)}{\partial c_i}. \qquad (c) \qquad (12.154)$$

If we consider Eq. (12.154(a)) for the moment for fixed i, we see that this is a constant-coefficient difference equation in the variable $\partial e(k)/\partial a_i$ with $y(k-i)$ as a forcing function. If we take the z-transform of this system and call $\partial e/\partial a_i \triangleq e_a$, then we find

$$E_{a_i}(z) = z^{-i} Y(z) - \sum_{j=1}^{n} c_j z^{-j} E_a(z)$$

or

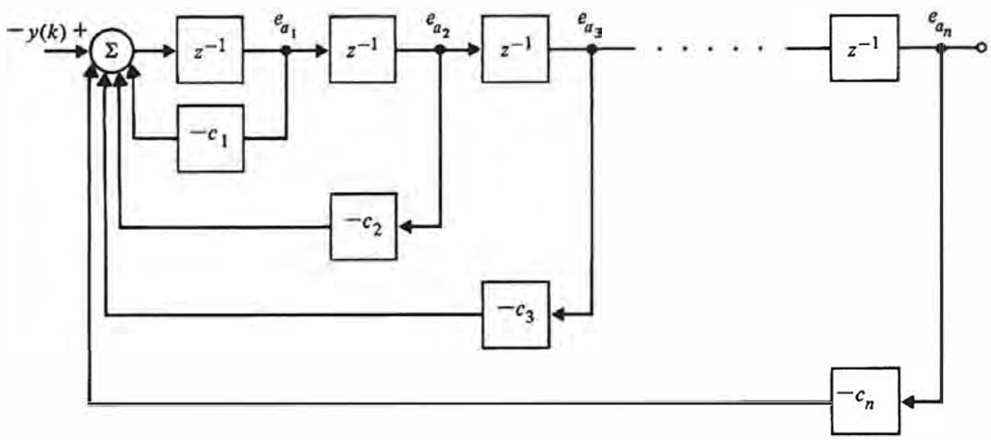
$$E_{a_i}(z) = \frac{z^{-i}}{1 + \Sigma_{j=1}^{n} c_j z^{-j}} Y(z).$$

Thus the derivative of e with respect to a_2 is simply z^{-1} times the partial derivative of e with respect to a_1, and so on. In fact, we can realize all of these partial derivatives via the structure shown in Fig. 12.9.

In exactly analogous fashion, dynamic systems, whose states are the sensitivities of e with respect to b_i and c_i, can be constructed. With these then, we have all the elements of an explicit algorithm that can be used to compute improvements in $\boldsymbol{\theta}$. The steps can be summarized as follows:

1. Select an initial parameter estimate, $\hat{\boldsymbol{\theta}}(0)$, based on analysis of the physical situation, the system step and frequency responses, cross-correlation of input and output, and/or least squares.
2. Construct (compute) $e(k)$ from Eq. (12.139) with e substituted for v; compute the sensitivities from Eq. (12.154) using three structures as shown in Fig. 12.9; and simultaneously compute $\hat{R}_v \partial \ell/\partial \boldsymbol{\theta}$ from Eq. (12.152) and the first term of $\hat{R}_v \partial^2 \ell/\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}$ from Eq. (12.153).
3. Compute $\hat{R}_v$ from Eq. (12.146) and solve for $\mathbf{g}$ and $\mathbf{Q}$.
4. Compute $\hat{\boldsymbol{\theta}}(K+1) = \hat{\boldsymbol{\theta}}(K) - \mathbf{Q}^{-1}\mathbf{g}$.
5. If $[\hat{R}_v(k+1) - \hat{R}_v(K)]/\hat{R}_v(K) < 10^{-4}$, stop. Else go back to Step 2.

Figure 12.9
Block diagram of dynamic system whose states are the sensitivities of e with respect to a_i



In Step 5 of this algorithm it is suggested that when the sum of squares of the prediction errors, as given by $\hat{R}_v$, fails to be reduced by more than a relative amount of 10^{-4}, we should stop. This number, 10^{-4}, is suggested by a statistical test of the significance of the reduction. A discussion of such tests is beyond our scope here but can be found in Astrom and Eykhoff (1971), and Kendal and Stuart (1967). If the order of the system, n, is not known, this entire process can be done for $n = 1, 2, 3, \ldots,$ and a test similar to that of Step 5 can be used to decide when further increases in n are not significant.

◆ **Example 12.5** *Identification of the Four Disk System*

To illustrate parameter identification with a non-trivial example, we consider the four disk mechanical set-up sketched in Fig 12.10 as studied by M. D. Sidman (1986). This is a representation of a physical system constructed in the Aeronautic Robotics Laboratory at Stanford University as part of a continuing study of the control of flexible mechanisms. In this case, the assembly has a torque motor coupled to disk 3 and angle sensors at disk 3 and at disk 1. The sensor at disk 3 is therefore collocated with the torque source and the sensor at disk 1 is non-collocated with the torque. All disks have the same inertia but, to permit parameter changes, provision is made to reduce the inertia of disk 4 by a factor of 0.5 or 0.25. Careful measurement of the transfer function by a one-frequency-at-a-time technique for each of the inertias of the disk was done and the magnitudes are plotted in Fig. 12.11. The problem is to estimate the transfer function by stochastic least squares.

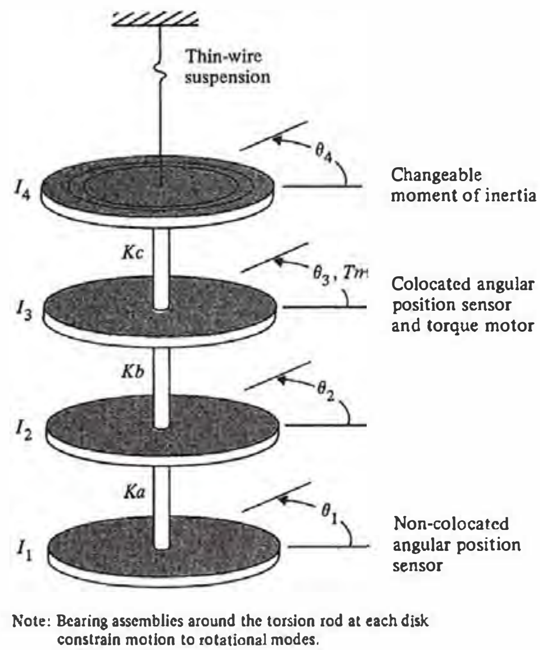


Figure 12.10
A four disk system, from Sidman (1986)

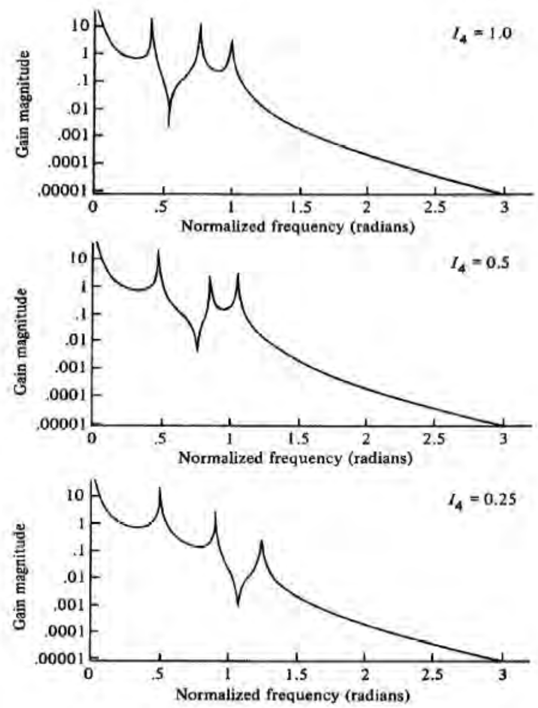


Figure 12.11
Frequency response for the four disk system measured with no noise and non-collocated signals, from Sidman (1986)

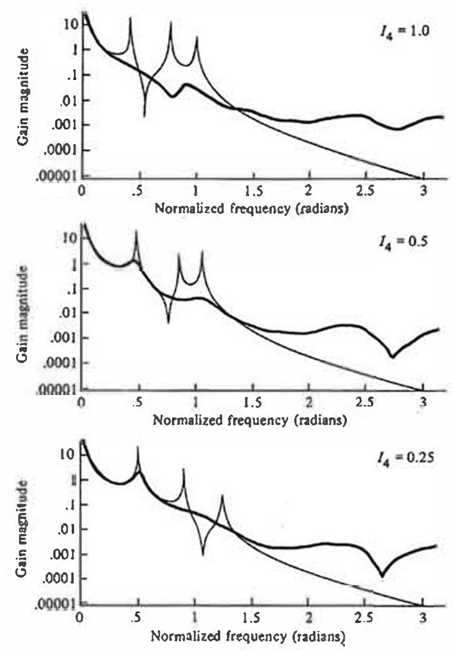


Figure 12.12 Frequency response of the four disk system found by least squares, from Sidman (1986)

Solution. If we write the $\theta_1(z)/U(z)$ transfer function as a ratio of two eight-degree polynomials, excite the system with a random signal, compute the least squares estimates of these parameters, and plot the resulting experimental frequency response, we obtain the curves shown in Fig. 12.12.

The poor fit obtained by using least squares in the presence of very small data converter quantization noise, W_a, at the plant output is obvious in Fig. 12.12. Because the flexible modes are not successfully identified, the estimates are worthless for the purpose of designing a controller that actively dampens these modes. The poor performance of least squares in

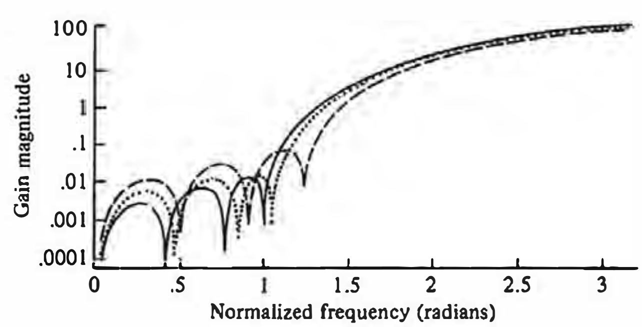


Figure 12.13 Frequency weighting caused by the system denominator, a(z), from Sidman (1986)

this situation is caused by the fact that the equation error is not white. It is colored by the characteristic polynomial of the plant, $a(z)$. Suppose we write the transfer function

$$Y(z) = \frac{b(z)}{a(z)} U(z) + W_a \tag{12.155}$$

Then the equation error is

$$a(z)Y(z) - b(z)U(z) = a(z)W_a. \tag{12.156}$$

Because $a(z)$ is typically large at high frequencies, and amplifies the noise, the parameter estimates will be greatly distorted by the least squares identification. This has the effect of corrupting the frequency response estimates as shown in Fig. 12.12. In the present case, a plot of the true $a(z)$ is shown in Fig. 12.13. If we filter the data with $a(z)$ as suggested by Eq. (12.156), we obtain the excellent results shown in Fig. 12.14. However, this polynomial is not known and cannot be used to reduce the high frequency distortion.

To reduce this effect, Sidman (1986) modified Clary's method and introduced the known parts and filtered the data with the structure shown in Fig. 12.15. The plant is divided into known and unknown parts, multiplied on both sides of Eq. (12.156) by the filter transfer function, F, and the new variables defined as

$$U_f(z) = U(z)\frac{b_k(z)}{a_k(z)} F(z), \tag{12.157}$$

$$Y_f(z) = Y(z)F(z). \tag{12.158}$$

Then the filtered least-squares formulation

$$a_u(z)[a_k(z)Y(z)] - b_u(z)[b_k(z)U(z)] = a_k(z)a_u(z)W_a \tag{12.159}$$

reduces to the modified formulation

$$a_u(z)Y_f(z) - b_u(z)U_f(z) = a_u(z)F(z)W_a \tag{12.160}$$

From these equations, one estimates the transfer function polynomials a_u and b_u of the unknown plant. However, as can be seen by comparing the right-hand sides of Eq. (12.159) and Eq. (12.160), this eliminates the coloration of the equation error due to the known a_k and introduces a filtering of the noise that can be used to reduce the weighting due to a_u in the estimates. In this example, a_k is a very significant cause of coloration at high frequencies since it includes the action of two roots at $z = 1$ corresponding to the plant's rigid body poles.

The bandpass filter $F(z)$ is selected to have a passband that brackets the region where the resonances are known to exist. This frequency constraint prefilter further reduces the effect of high and low frequency noise and disturbances including broadband quantization noise, DC bias in the sensor and actuator, low-frequency torque disturbances, as well as excited unmodeled dynamics above the frequency band of interest. The high pass portion of the bandpass filter may be chosen to have its zeros at $z = 1$. This leads to a stable, simple combined prefilter that lacks undesirable low frequency behavior. In this example, b_k consists of real zeros located on the negative real axis. The contribution to the system response due to the zeros of b_k is very insensitive to changes in the inertia or torsional stiffness in the plant. Thus, the identification need only estimate one complex zero pair and three complex pole pairs of the four disk system.

With the suggested pre-processing of the input and output data, the results of Fig. 12.16 are obtained. This model is quite suitable for use in control or in adaptive control schemes.

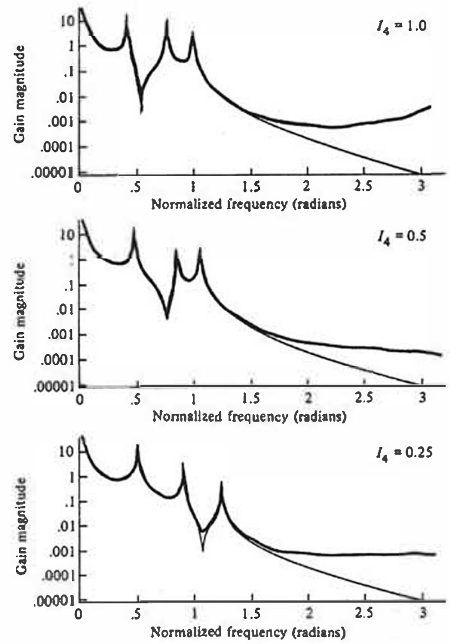


Figure 12.14
Frequency response using filtered least squares, from Sidman (1986)

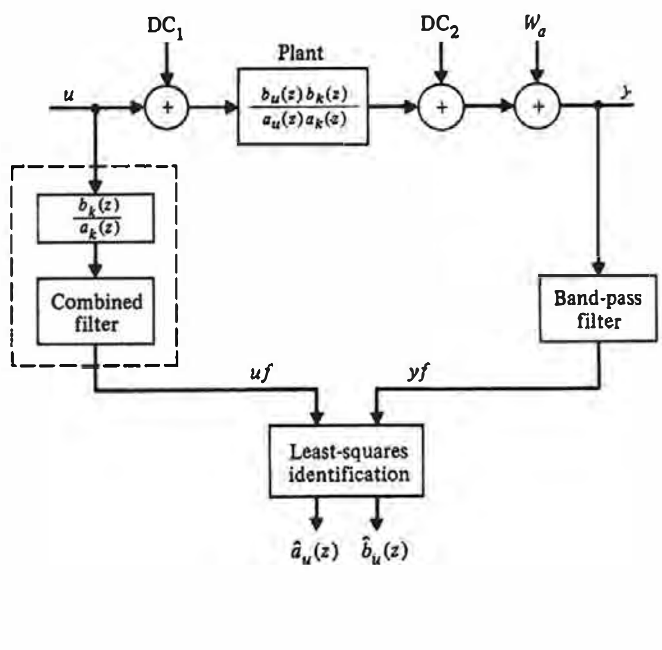


Figure 12.15
Structure of modified filtered least squares with frequency constraint prefiltering, from Sidman (1986)

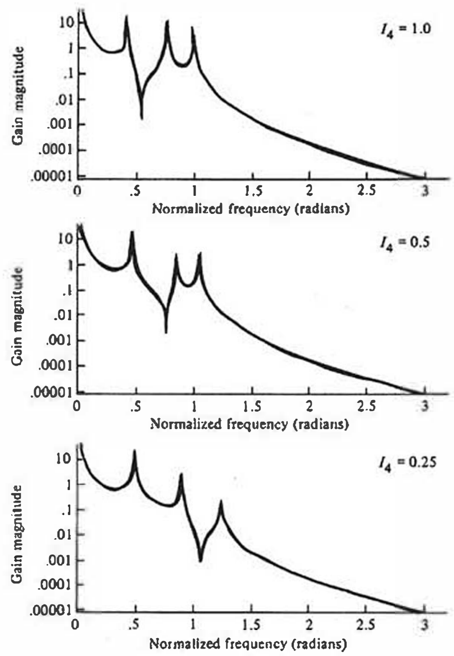


Figure 12.16
Frequency response of four disk system using least squares with low-pass filters and known factors, from Sidman (1986)

12.8 Subspace Identification Methods

The identification methods described thus far have been based on transfer function models. The nonparametric methods estimate the transfer function directly as a function of frequency and the parametric methods are based on representing the input-output relation as a rational function with unknown coefficients for the numerator and denominator polynomials. An alternative formulation of the identification problem is to express the input-state-output relationships in state space form with unknown description matrices $\mathbf{\Phi}$, $\mathbf{\Gamma}$, $\mathbf{H}$, and J. In addition to these matrices, the (unknown) state sequence, $\mathbf{x}(k)$, is introduced as well. At first blush, it would seem that this structure greatly complicates the problem as there are many more parameters to be considered as compared to the transfer function model. The approach is to express the data in an *expanded* space guaranteed to contain the state description and to extract the *subspace* consistent with the state equations using reliable and robust methods of numerical linear algebra. The methods have been to shown to have very desirable statistical properties and, perhaps most important, to be directly applicable to multivariable systems with vector inputs and outputs. The subspace-based methods have their origins in the early work toward finding a state realization from a deterministic impulse response as described in Ho (1966). A unification of the field has been given in Overshee (1995) where the method described here (based on Cho (1993)) is called a prediction method.

The method begins with a collection of input and output data $u(k)$ and $y(k)$ that we assume are scalars but could just as well be vectors for this method. To motivate the development of the algorithm, it is assumed that this data comes from a noise-free underlying state system given by equations of the form

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{\Phi} x(k) + \mathbf{\Gamma} u(k) \\ y(k) &= \mathbf{H}\mathbf{x}(k) + J u(k) \end{aligned} \tag{12.161}$$

where the (unknown) dimension of the state is n. In order to express the constraints imposed on the input and output sequences by Eq. (12.161), a number, M, known to be larger than the state dimension but much smaller than the data length, is selected and the data are organized into the matrices $\mathbf{Y}$ and $\mathbf{U}$ of size $M \times N$ as follows

$$\mathbf{Y} = \begin{bmatrix} y_0 & y_1 & \cdots & y_{N-1} \\ y_1 & y_2 & \cdots & y_N \\ \vdots & \vdots & \vdots & \vdots \\ y_{M-1} & y_M & \cdots & y_{N+M-2} \end{bmatrix}. \tag{12.162}$$

and

$$\mathbf{U} = \begin{bmatrix} u_0 & u_1 & \cdots & u_{N-1} \\ u_1 & u_2 & \cdots & u_N \\ \vdots & \vdots & \vdots & \vdots \\ u_{M-1} & u_M & \cdots & u_{N+M-2} \end{bmatrix}. \tag{12.163}$$

We assume that the input signals are "exciting" in the sense that $u(k)$ contains many frequencies and, in consequence, the matrix $\mathbf{U}$ has full rank $= M$. To these data matrices we add the $n \times N$ matrix of states

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_0 & \cdots & \mathbf{x}_{N-1} \end{bmatrix} \tag{12.164}$$

Finally, we define composite matrices made from the description matrices as

$$\mathcal{O}_x = \begin{bmatrix} \mathbf{H} \\ \mathbf{H\Phi} \\ \vdots \\ \mathbf{H\Phi}^{M-1} \end{bmatrix} \tag{12.165}$$

and

$$\mathcal{H} = \begin{bmatrix} J & 0 & 0 & 0 & 0 \\ \mathbf{H\Gamma} & J & 0 & 0 & 0 \\ \mathbf{H\Phi\Gamma} & \mathbf{H\Gamma} & J & 0 & 0 \\ \vdots & \cdots & \cdots & \ddots & \vdots \\ \mathbf{H\Phi}^{M-2}\mathbf{\Gamma} & \mathbf{H\Phi}^{M-3}\mathbf{\Gamma} & \cdots & \cdots & J \end{bmatrix}. \tag{12.166}$$